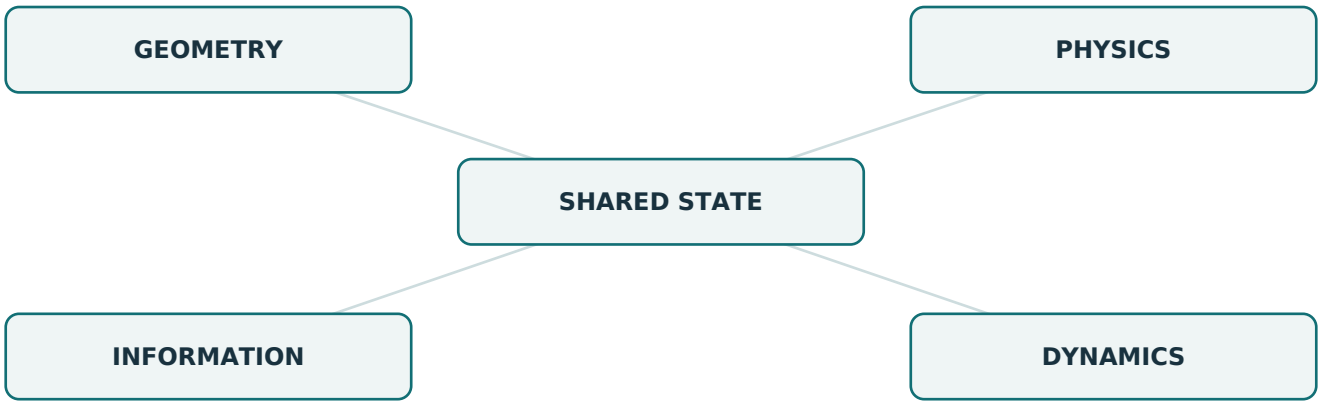
XD-TAU

An Integrated Architecture of Geometry, Physics and Information

Binary-spherical organization.
Continuous fields. Coupled physical domains.

A formal expansion of the Binary-Spherical Organogram Architecture through constructive geometry, material dynamics, transport, control, graphs, coding and spacetime.



EXPANDED DOMAINS / CORRECTED SEMANTICS / PROVED PROFILES

Tom Klootwijk

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The Diff-Force-CSG expansion, with explicit physical models, constructive proofs and reproducible reference updates.

SYNTHESIS / 02

The expanded domain map

XD-TAU coordinates addressable entities, implicit geometry and evolving fields through shared state. The names below identify formal representations or specialized adapters, each defined by equations in this edition.

Domain	Formal connection	Pages
Geometry / CAD	Functional representation; regularized CSG; differentiable parameter fitting	4-6
Topology	Klein immersion, thickened image and level-set evolution	7
Geospatial computing	Binary equal-area hierarchy, zonal lookup and spherical metrics	8-9
Information visualization	Log-polar hierarchy and Poincaré focus/context geometry	10
Analog coordinate design	Shark profile, geyser map, barycentric chart and spiral phase	11
Continuum mechanics	Control volumes, material deformation, stress and traction	12-13
Fluid / thermal physics	Mass, momentum, energy, plume transport and drag-to-heat transfer	14-16
Optics	Absorption, radiative source coupling and material response	16
Control / cybernetics	Port-Hamiltonian coupling and self-referential state evolution	6, 17, 20
Graphs / neuroscience	Laplacian diffusion, neural fields and connectivity signals	18
Hardware / data structures	Packed topology, rank/select, XOR kernels and propagation time	19
Distributed systems	Join-semilattice operation logs and replicated state	20
Coding / quantum information	Linear parity, graph-state stabilizers and nonlinear Boolean channels	21-22
Spacetime geometry	3+1 metric, lapse, shift and ADM state	23
Orbital / network systems	Ephemerides, directional cells, control and beamforming	24

Pages 25-26 connect the domains. The completion chapters add proofs, domains, numerical maps, initial data and the ADM system. Separate records identify the earlier 27 formula checks and the new exact, symbolic and floating-point verifications.

FOUNDATION / 03

State, time and dimensional meaning

$$s_n = (\mathsf{E}_n, \mathsf{H}_n, P_n, \xi_n, b_n, \mathsf{F}_n, \mathsf{O}_n, \mathcal{C}_n) \tag{1.1}$$

E is the entity table; H the address hierarchy; P the parenthesized geometry program; ξ the continuous parameters; b the packed bits; F the physical and graph fields; O the tagged operation set; and C the declared couplings. Each module reads the current state and contributes to the next state.

$$e = (\mathrm{ID}, \mathrm{orgParent}, w, \mathbf{x}, \mathrm{payload}) \tag{1.2}$$

An ID identifies an entity. The binary word w identifies a spatial region. Organizational parents, spatial coordinates and interaction edges are distinct stored relations. A subtree query resolves entity IDs before resolving their distributed positions.

One clock, several time coordinates

$$\tau_\circ = 2\pi, \quad n \in \mathbb{N}, \quad T = \ln(1 + t/t_*), \quad t_* > 0,\ t \geq 0 \tag{1.3}$$

$$t = t_*(e^T - 1), \qquad \partial_T = (t + t_*)\partial_t \tag{1.4}$$

The full turn τ_circle is angular. n is the update index. t is physical time, and T is its dimensionless logarithmic coordinate. Physical increments follow:

$$\Delta t_n = t_*(e^{T_{n+1}} - e^{T_n}) \tag{1.5}$$

Quantity	Domain and units
Position, radius, exact distance	Euclidean R^3 or a declared manifold; metres
Field Φ and blend width h	Compatible scalar-field units; length for distance-valued primitives
Velocity, force, energy	m/s, N and J in a physical adapter
Density, pressure, temperature	kg/m³, Pa and K
Optimization state ξ and loss L	Nondimensional coordinates with stored parameter scales
Bits, masks and adjacency	Finite binary vectors and matrices over F_2

Domain definitions specify the inputs for each equation. Couplings record units, coordinate frames, sample locations and time epochs, so a geometric parameter, an information bit and a physical force keep their respective meanings.

GEOMETRY / CAD / 04

Functional representation and full CSG

Geometry has an authoritative region and auxiliary scalar fields. Put X = R³, ρ(E) = cl(int E), and RC(X) = {A : A = ρ(A)}. Default import of any finite field, including zero plateaus and discontinuous masks, is:

$$\mathcal{I}(f) = \rho\{x : f(x) < 0\}, \qquad A \in \mathrm{RC}(X) \tag{2.1}$$

$$A \vee B = A \cup B, \qquad A \wedge B = \rho(A \cap B) \tag{2.2}$$

$$A^{*} = \mathrm{cl}(X \setminus A), \qquad A \setminus_{*} B = A \wedge B^{*} \tag{2.3}$$

Import leaves first, then evaluate these region operators. Each exact node stores its resulting region. The zero and unit are the empty region and X. Lower-dimensional contacts are stored in a separate feature channel. The Boolean-algebra proof appears in the completion chapters [R22].

Metric and finite field semantics

$$d_A(x) = \mathrm{dist}(x, A) - \mathrm{dist}(x, X \setminus A) \tag{2.4}$$

$$d_{\varnothing} = +\infty, \qquad d_X = -\infty \tag{2.5}$$

$$\phi_A = \frac{\ell d_A}{\ell + |d_A|} \quad (\varnothing \neq A \neq X), \qquad \phi_{\varnothing} = \ell, \quad \phi_X = -\ell \tag{2.6}$$

Here ell > 0 is a length. For proper regions, d_A and φ_A are spatially 1-Lipschitz; their negative, zero and positive values identify interior, boundary and exterior. A = I(φ_A) = {φ_A ≤ 0}. Occupancy q_A(x)=1_A(x) includes the boundary; empty/full occupancy is 0/1.

$$\mathcal{I}(\min(\phi_A, \phi_B)) = A \vee B, \quad \mathcal{I}(\max(\phi_A, -\phi_B)) = A \setminus_{*} B \tag{2.7}$$

Hard field formulas in (2.7) consume canonical inputs. Arbitrary raw fields with the same negative region need not preserve these identities under raw negation or subtraction. A field type records membership, distance bound or exact signed distance; the auxiliary fitting field remains available.

DIFFERENTIABLE GEOMETRY / 05

Smooth CSG and analytic derivatives

The original 45-degree encircled blend becomes the smooth-min operator S_h. Its dual T_h supplies smooth intersection and subtraction. Let a, b and h carry the same field units, with h > 0.

$$D_h = \sqrt{(a-b)^2 + 2h^2} \tag{3.1}$$

$$S_h(a,b) = \tfrac{a+b-D_h}{2}, \qquad T_h(a,b) = \tfrac{a+b+D_h}{2} \tag{3.2}$$

$$\Phi_{\text{union}} = S_h(a,b), \quad \Phi_{\text{intersection}} = T_h(a,b), \quad \Phi_{\text{difference}} = T_h(a,-b) \tag{3.3}$$

$$\partial_a S_h = \tfrac{1}{2}\left(1 - \tfrac{a-b}{D_h}\right), \quad \partial_b S_h = \tfrac{1}{2}\left(1 + \tfrac{a-b}{D_h}\right) \tag{3.4}$$

$$\partial_h S_h = -h/D_h, \qquad \partial_h T_h = h/D_h \tag{3.4b}$$

$$0 < \min(a,b) - S_h(a,b) \le h/\sqrt{2} \tag{3.5}$$

Equation (3.5) bounds scalar change. Smooth nodes use canonical distance inputs; tagged infinite inputs are evaluated by the min/max limits. Smooth operations are geometric deformations and are nonassociative and nonidempotent. The completion chapters give the spatial bound and the final-field derivative rule. Chain rules hold where the composed inputs are differentiable.

Normals and learned leaves

$$\mathbf{n} = \tfrac{\nabla\Phi}{\|\nabla\Phi\|}, \qquad \Phi = 0, \quad \nabla\Phi \neq 0 \tag{3.6}$$

$$f_\theta(\mathbf{x}) = f_{\text{analytic}}(\mathbf{x}) + r_\theta(\mathbf{x}) \tag{3.7}$$

A learned residual r_θ or a neural implicit leaf supplies the neural-geometry specialization. The leaf stores network parameters, input normalization and a fitting objective. Analytical and learned leaves share the same constructive tree [R3].

An image-fitting adapter may use differentiable rasterization, including the primitive-intersection gradients described by DiffCSG [R2]. Field fitting uses the analytic derivatives above.

DYNAMICS / CYBERNETICS / 06

Optimization force and direct evolution

Let ξ be nondimensional continuous parameters, P the geometry expression and b the discrete state. Store dimensional parameters as $\theta = \theta_{\text{scale}} \odot \xi$. A fitting objective defines an optimization direction.

$$\hat{q}_\xi(\mathbf{x}) = \sigma(-\Phi_{\xi,P}(\mathbf{x})/\varepsilon), \quad \sigma(z) = (1+e^{-z})^{-1}, \quad \varepsilon > 0 \tag{4.1}$$

$$L(\xi) = \frac{1}{N}\sum_i (\hat{q}_\xi(\mathbf{x}_i) - y_i)^2, \qquad \xi_{n+1} = \xi_n - \eta G_\xi^{-1} \nabla_\xi L \tag{4.2}$$

G_ξ is positive definite and $\eta \geq 0$. Use the ordinary objective gradient where locally C^1 and the final-objective generalized selection elsewhere on its locally Lipschitz domain. The vector $-G_\xi^{-1}\nabla L$ is the optimization "force." Mechanical forces enter the physical momentum equation in newtons.

Continuous, discrete and topology channels

$$b_{n+1} = b_n \oplus u_n, \qquad P_{n+1} = \text{Edit}(P_n, e_n) \tag{4.3}$$

$$s_{n+1} = \mathcal{U}_h(s_n, e_n), \quad B_{n+1} = \text{Encode}(Q_\delta(s_{n+1})) \tag{4.4}$$

The update reads one state snapshot. Continuous fitting, bit mutations, physical field steps, entity events and expression edits produce the next state directly. The completion chapters define a causal execution order, concrete couplings and a finite reference implementation. Occupancy samples and continuous field derivatives are available together.

Local automata and self-reference

$$b_i^{n+1} = u_\theta(\{b_j^n : j \in \mathcal{N}(i)\}, q_i^n), \qquad \theta^{n+1} = v(\theta^n, b^n, q^n) \tag{4.5}$$

Uniform local rules give a graph cellular automaton. Including θ and the geometry program in the encoded state gives a self-referential geometric state machine. A boundary-regenerating module is specified by an explicit production rule that rebuilds its defining region from its own products; this supplies the constructive autopoietic specialization.

Coupling C maps declared outputs into other modules' inputs. Display deformations, geometry and physical fields may participate in feedback through those explicit couplings.

CONTINUOUS GEOMETRY / 07

Topology, thickened images and interfaces

Let K be the abstract Klein bottle and ι:K→R³ a selected immersion. Its image S is a spatial set. Thickness ε gives a volumetric representation while preserving the distinction between abstract topology and the image geometry.

$$S = \iota(K), \qquad \Omega_{K,\varepsilon} = \{\mathbf{x} : \operatorname{dist}(\mathbf{x}, S) \le \varepsilon\}, \quad \varepsilon > 0 \tag{5.1}$$

$$f_\varepsilon(\mathbf{x}) = \operatorname{dist}(\mathbf{x}, S) - \varepsilon, \qquad d_K = d_{\Omega_{K,\varepsilon}} \tag{5.2}$$

f_ε is an offset membership field. d_K is the derived signed distance. Self-intersections become overlaps in the image's thickened volume. The program stores the immersion, scale and thickness as reproducible parameters.

An evolving interface

$$\partial_t \Phi + \mathbf{v} \cdot \nabla \Phi = 0 \tag{5.3}$$

$$\partial_t \Phi + V_n \|\nabla \Phi\| = 0 \tag{5.4}$$

Advection by velocity v and normal motion with speed V_n provide the Eulerian level-set connection [R4]. V_n can be supplied by curvature, pressure, a fitting direction or a material law. The finite field transports the interface; occupancy and distance are derived outputs.

Geometry, topology and physics

A geometric shell uses a selected solid thickness. A physical shell additionally carries density, stiffness, temperature and material motion. The mechanical adapter maps these fields to stress and traction. Geometric connectivity can change through CSG events; physical interface evolution follows the chosen field equations.

$$\Phi_n = S_h(\min(f_{\text{pyr}}, f_{\text{cone}}), \min(f_{\text{sphere}}, m_n f_K)), \quad m_n(\mathbf{x}) = M_{i_n(\mathbf{x}), j_n(\mathbf{x})} \tag{5.5}$$

Equation (5.5) is the source's auxiliary accumulated field. Its discontinuous mask is evaluated per cell and imported through I. Exact Boolean nodes consume canonical regions. Seam jumps have the weak derivative given in the completion chapters.

> **Formal connection**
>
> The Klein component is an immersed-surface image with a volumetric offset. The level-set component is an evolving implicit interface. Together they provide topology-aware geometric state, with material behavior supplied by the mechanics and transport adapters.

GEOSPATIAL DOMAIN / 08

A concrete binary global grid

Use u = y/R and longitude ψ about the y-axis. The spherical map and its area element are:

$$\mathbf{p}(u,\psi) = R(\sqrt{1-u^2}\cos\psi, u, \sqrt{1-u^2}\sin\psi) \tag{6.1}$$

$$-1 \le u \le 1, \quad 0 \le \psi < 2\pi, \qquad dA = R^2\,du\,d\psi \tag{6.2}$$

Start with $[-1,1]\times[0,2\pi)$. Odd address bits bisect the u interval; even bits bisect the ψ interval. A split boundary belongs to the upper child. The first bit therefore reproduces the original y = 0 hemispherical cut. At either pole, use ψ = 0.

$$n_u = \lceil d/2 \rceil, \quad n_\psi = \lfloor d/2 \rfloor, \quad \mu(C_w) = 4\pi R^2/2^d \tag{6.3}$$

$$j(w) = \sum_{i=1}^{d} b_i 2^{d-i}, \qquad \mathrm{addr}(w) = 2^d - 1 + j(w) \tag{6.4}$$

Every level gives equal-area cells. Midpoint subdivision of u and ψ refines both directions. Pole and seam ownership is unique, with area-zero boundary assignments.

Representatives and metric size

$$\widehat{\mathbf{p}}_w = \mathbf{p}(u_c, \psi_c), \quad u_c = (u_- + u_+)/2, \quad \psi_c = (\psi_- + \psi_+)/2 \tag{6.5}$$

$$\varepsilon_w = R\left[\max_{u\in\{u_-,u_+\}} |\arcsin u - \arcsin u_c| + \tfrac{\psi_+ - \psi_-}{2}\right] \tag{6.6}$$

A meridian-then-parallel path gives d_S(p,p̂_w) ≤ ε_w. Both interval widths decrease with depth, so this radius bound tends to zero. The cell area and radius are metadata for aggregation and resolution selection.

The zonal API supplies locate, cell, representative, parent, children, neighbors and aggregate. Neighbors are cells whose closures share a boundary; cross-seam and polar contacts follow the same definition. This provides the explicit global-grid connection [R5].

SPHERICAL NAVIGATION / 09

OTAN2, coordinate recovery and metrics

The original hemispherical chart remains available beside the equal-area grid. Let h be the y hemisphere, $s_h = 2h-1$ and $r = \sqrt{(x^2+z^2)}$.

$$\rho = \ln(1 + r/r_*), \quad r_* > 0, \qquad 0 \le \rho \le \ln(1 + R/r_*) \tag{7.1}$$

$$q = \mathrm{frac}\left(\frac{\mathrm{atan2}(z, x) + \omega_n}{2\pi}\right), \quad \omega_n = [\omega_0 + \nu T_n]_{2\pi} \tag{7.2}$$

$$r = r_*(e^{\rho} - 1), \qquad \psi = [2\pi q - \omega_n]_{2\pi} \tag{7.3}$$

$$\mathbf{p} = (r\cos\psi, s_h\sqrt{R^2 - r^2}, r\sin\psi) \tag{7.4}$$

The same phase epoch is used for both directions. The seam belongs to h = 1; poles use one designated angular code. The return map satisfies $\|p\|^2 = R^2$.

$$d_S(\mathbf{p}, \mathbf{q}) = R\,\mathrm{atan2}(\|\mathbf{p} \times \mathbf{q}\|, \mathbf{p} \cdot \mathbf{q}) \tag{7.5}$$

Entity records can store exact numerical coordinates or a cell representative plus a residual. Coordinate precision, field sampling resolution and spatial cell size are separate metadata. The regularized radius keeps ρ finite at the poles.

Domain-specific distances

Domain	Metric or geometric object
Sphere navigation	Geodesic distance (7.5) and cell radius ε_w
Solid geometry	Euclidean boundary distance d_Ω
Hyperbolic hierarchy	Poincaré metric on the unit disk
Material mechanics	Reference/deformed metrics and strain
Optimization	Parameter metric G_ξ
Spacetime	Lorentzian line element with lapse and shift

The lookup layer carries the metric type with its data. The pinion supplies angular phase; recorded movement supplies position changes.

INFORMATION VISUALIZATION / 10

Log-polar and hyperbolic organization

A binary hierarchy retains the original L-system production N(w)→N(w0)N(w1). Its log-polar layout separates displayed depth from geographic position.

$$\theta_w = \frac{2\pi(j(w)+1/2)}{2^d}, \quad r_d = r_0 a^d, \quad \rho_d = d\ln a, \quad a > 1 \tag{8.1}$$

Depth is linear in log radius and angular sectors halve at each branch. Entity IDs link this layout to the organizational parent graph and spherical records.

The Poincaré focus/context adapter

$$z_w = \tanh(\kappa d/2)e^{i\theta_w}, \quad \kappa > 0, \qquad ds^2 = \frac{4|dz|^2}{(1-|z|^2)^2} \tag{8.2}$$

$$d_H(0, z_w) = 2\operatorname{artanh}|z_w| = \kappa d \tag{8.3}$$

$$F_a(z) = \frac{z-a}{1-\bar{a}z}, \qquad |a| < 1, \ |z| < 1 \tag{8.4}$$

The Poincaré disk supplies exponential spatial capacity with a finite circular display. F_a moves focus a to the center while preserving hyperbolic distances. This gives the formal focus/context connection [R6].

Hierarchy, interaction and identity

The organizational relation can be nonbinary. Display grouping nodes give a binary layout while original parents remain in the entity records. Interaction graphs can include cycles and weighted edges. Selecting a display node resolves its entity set; moving focus changes the chart coordinates.

A conformal map preserves local angles where its complex derivative is nonzero. The disk automorphism in (8.4) gives a concrete conformal operation. The logarithmic radial layout and the hyperbolic renderer are two explicitly defined display choices.

> **Coupling**
>
> Layout coordinates may drive interaction, sampling or geometric parameters through a declared C map. The selected coupling records which quantities change, while IDs keep the corresponding entities identifiable.

ANALOG GEOMETRY / 11

Shark-Geyser, barycentric and spiral maps

$$f_p(x) = |(x \bmod p) - p/2|, \quad p > 0 \tag{9.1}$$

$$r' = r_0 e^{\alpha_g (r - r_c)}, \quad r_0 > 0, \qquad r = r_c + \alpha_g^{-1} \ln(r'/r_0) \tag{9.2}$$

The triangular profile supplies a periodic folded amplitude. Retaining its period and branch indices gives an inverse description. The geyser map has an inverse for $\alpha_g \neq 0$; α_g carries inverse-length units. A chamber function supplies the accumulation stage.

$$\mathcal{K}(r) = r_c + \ell \tanh((r - r_c)/\ell), \qquad r_c \geq \ell > 0, \quad r \geq 0 \tag{9.2b}$$

This canonical chamber compresses radial increments smoothly; the eruption takes its output as the input radius.

$$\lambda_1 + \lambda_2 + \lambda_3 = 1, \quad \mathbf{u} = \sum_i \lambda_i \mathbf{v}_i, \quad \rho_{\text{raw}} = \ln(r/r_*), \ r > 0 \tag{9.3}$$

$$s = \rho/\rho_{\max}, \quad t = \psi/(2\pi), \quad (\lambda_1, \lambda_2, \lambda_3) = (1 - s, s(1 - t), st) \tag{9.3b}$$

$$\text{Len} = s + \gamma_f 2 f_p(s)/p, \qquad \gamma_f \geq 0 \tag{9.3c}$$

The canonical chart uses ρ from (7.1), $\rho_{\max} = \ln(1+R/r_*)$, and a nondegenerate triangle. For $s > 0$, recover $t = \lambda_3/(\lambda_2+\lambda_3)$ and $s = \lambda_2+\lambda_3$. Radial edges identify the seam; $s = 0$ uses the pole code. Nonnegative weights give triangle containment; affine weights also describe its exterior. In Len, p and γ_f are dimensionless.

$$\Theta_{\text{strip}}(\theta) = \theta + \alpha_\theta \tanh((\theta - \beta)/\sigma) \sin(2\pi \text{Len}), \quad \sigma > 0 \tag{9.4}$$

$$\Theta_{\text{circle}}(\theta) = \theta + a \sin(\theta - \beta), \qquad |a| < 1 \tag{9.5}$$

The strip map retains the source's angular profile. The circle map adds a periodic, strictly increasing phase adapter: $\Theta(\theta+2\pi) = \Theta(\theta)+2\pi$ and $\Theta' > 0$. β is a selected axis direction with a stored frame and sign convention.

$$z = k\Theta_{\text{unwrapped}}, \qquad I(r) = \sin(\omega_s \ln(r/r_*)), \quad r > 0 \tag{9.6}$$

$$I(qr) = I(r), \qquad q = e^{2\pi/|\omega_s|}, \quad \omega_s \neq 0 \tag{9.7}$$

The unwrapped phase retains axial displacement across turns. k has length-per-radian units. The log-sine has exact discrete scale repetition. A chosen interval $[r_{\min}, r_{\max}]$ specifies the sampled scales and cycle count for a particular rendering or physical coefficient field.

CONTINUUM DISCRETIZATION / 12

Surface cells, volume cells and flux

Surface quantities live on S^2 or its cell graph. Three-dimensional transport uses radial layers as well as angular cells. Let C_w be a cell on the unit sphere, with solid angle ω_w.

$$K_{w,k} = \{r\mathbf{s} : \mathbf{s} \in C_w, \quad r_k \le r < r_{k+1}\} \tag{10.1}$$

$$|K_{w,k}| = \frac{\omega_w}{3}(r_{k+1}^3 - r_k^3) \tag{10.2}$$

Store face areas, outward normals and neighbor incidence. Shared faces use the same geometric measure and opposite orientations.

$$|K_i|U_i^{n+1} = |K_i|U_i^n - \Delta t \sum_{f \subset \partial K_i} |f|\widehat{F}_{i,f} + \Delta t|K_i|S_i \tag{10.3}$$

U contains cell-averaged conserved variables. F_{hat} is the numerical flux and S is the source. Each internal face contributes equal and opposite transfers to its two cells.

$$\sum_i |K_i|(U_i^{n+1} - U_i^n) = -\Delta t \sum_{f \in \partial \mathcal{D}} |f|\widehat{F}_f + \Delta t \sum_i |K_i|S_i \tag{10.4}$$

Internal cancellation establishes the conservative finite-volume connection. The remaining change is boundary transport plus physical sources. This balance is built into the evolution formula itself.

Surface transport

$$\partial_t c + \nabla_S \cdot (c\mathbf{v}_S) = D\Delta_S c + s \tag{10.5}$$

Here c is an areal density, with tangent velocity v_S, surface divergence and the Laplace-Beltrami operator. This supports population density, ecological transport and spatial signal fields. Surface temperature uses an areal heat-capacity balance. Volume and surface adapters share identities while storing their respective physical measures.

> **Formal connection**
>
> The binary hierarchy supplies addresses and aggregation. Control-volume geometry supplies physical areas and volumes. Antisymmetric shared-face transfers supply conservation.

MECHANICS / 13

Material motion, stress and shells

A material point X moves to x = χ(X,t). The deformation gradient and Green-Lagrange strain define the mechanical meaning of a geometric warp.

$$F = \frac{\partial \chi}{\partial \mathbf{X}}, \quad J = \det F > 0, \quad E_G = \tfrac{1}{2}(F^T F - I) \tag{11.1}$$

$$\rho J = \rho_0, \qquad \rho_0 \ddot{\chi} = \operatorname{Div} P + \rho_0 \mathbf{b} \tag{11.2}$$

The orientation-preserving deformation domain has J > 0. ρ is current density and ρ₀ reference density. P is the first Piola-Kirchhoff stress, with units of force per reference area.

$$P = \frac{\partial W(F,\vartheta)}{\partial F}, \qquad \sigma = J^{-1} P F^T \tag{11.3}$$

W is Helmholtz free-energy density per reference volume, in J/m³; the derivative in (11.3) holds temperature fixed. For small strain ε = (∇u+∇uᵀ)/2, isotropic elasticity gives σ = λ tr(ε)I+2με. Temperature-dependent W supplies thermal expansion and other coupled responses [R11].

Implicit boundaries and pressure

$$\mathbf{n} = \nabla\Phi / \|\nabla\Phi\|, \qquad \mathbf{t} = \sigma \mathbf{n}, \qquad \mathbf{t}_p = -p\mathbf{n} \tag{11.4}$$

Traction acts on regular implicit-surface points. A physical shell uses its thickness, material density, constitutive law and boundary loading. The same cone, sphere or thickened-image geometry can therefore author a mechanical domain.

Coordinate metric and material strain

$$G_\chi = (D\chi)^T D\chi \tag{11.5}$$

G_χ is a pullback metric. When χ maps material points, it measures physical deformation. When χ maps a display chart, it measures chart distortion. The coupling type records which interpretation applies.

The mechanics adapter contributes displacement, velocity, stress, traction and material energy to shared state. Geometry can supply its boundary, and the solved motion can update geometry parameters.

FLUID PHYSICS / 14

Fluid motion and thermodynamic state

Use density ρ, velocity v, pressure p, temperature ϑ and specific internal energy e on a three-dimensional domain. The conserved state is U = (ρ,ρv,ρE), with E = e+|v|²/2.

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = 0 \tag{12.1}$$

$$\partial_t (\rho \mathbf{v}) + \nabla \cdot (\rho \mathbf{v} \otimes \mathbf{v} + pI - \tau) = \rho \mathbf{b} \tag{12.2}$$

$$\partial_t (\rho E) + \nabla \cdot [(\rho E + p)\mathbf{v} - \tau \mathbf{v} + \mathbf{q}] = \rho \mathbf{b} \cdot \mathbf{v} + Q \tag{12.3}$$

τ is viscous stress, q heat flux, b body acceleration and Q supplied heat per volume per time. These are the continuum mass, momentum and total-energy balances [R7].

A concrete constitutive family

$$\tau = 2\mu D + \lambda_v (\nabla \cdot \mathbf{v}) I, \quad D = \tfrac{1}{2}(\nabla \mathbf{v} + \nabla \mathbf{v}^T), \quad \mathbf{q} = -k \nabla \vartheta \tag{12.4}$$

$$p = \rho R_s \vartheta, \qquad e = c_v \vartheta \tag{12.5}$$

The Newtonian ideal-gas family stores shear viscosity μ, second viscosity λ_v, conductivity k, gas constant R_s and specific heat c_v. Bulk viscosity is $\zeta = \lambda_v + 2\mu/3$. Other material models enter through their own stress, heat-flux and equation-of-state functions.

Plumes, convection and entrainment

Gravity and temperature-dependent density produce buoyancy. Velocity transports heat and concentration, pressure redirects the flow, viscosity converts mechanical energy into internal energy, and heat flux redistributes temperature. A plume’s visible envelope is a geometry output of this evolving state.

The finite-volume update on page 12 transports these conserved variables between volume cells. Boundary data and source fields supply inlet flow, ambient conditions, mechanical work and heating. Logical log-time schedules the physical increments through (1.4).

PLUME / ENERGY COUPLING / 15

Shiitake morphology, drag and heat

A cap-and-stem field gives the source's Shiitake morphology. For a toroidal cap and cylindrical stem aligned with z, one useful geometric profile is:

$$f_{\text{cap}} = \sqrt{(\sqrt{x^2+y^2}-a)^2+(z-z_c)^2}-b \tag{13.1}$$

$$f_{\text{stem}} = \max(\sqrt{x^2+y^2}-r_s, |z-z_s|-\ell_s), \quad \Phi_{\text{plume}} = S_h(f_{\text{cap}}, f_{\text{stem}}) \tag{13.2}$$

The cap radius a, tube radius b and stem dimensions are geometric parameters. These raw-field formulas define an auxiliary envelope. The metric bound (A.8) applies to blends of canonical regional distances. Density, velocity and temperature supply the physical plume state; a selected isovalue or fitted envelope links the two.

Reduced drag channel

$$\mathbf{w} = \mathbf{v}_{\text{body}} - \mathbf{v}_{\text{ambient}}, \qquad \mathbf{F}_D = -\tfrac{1}{2}\rho C_D A \|\mathbf{w}\|\mathbf{w} \tag{13.3}$$

$$P_{\text{diss}} = -\mathbf{F}_D \cdot \mathbf{w} = \tfrac{1}{2}\rho C_D A \|\mathbf{w}\|^3 \tag{13.4}$$

ρ is ambient density, C_D the dimensionless drag coefficient and A the reference area [R8]. Equal and opposite momentum transfers couple body and medium. The dissipated relative mechanical power supplies a heat channel. This reduced model represents unresolved drag.

Energy accounting

$$\tfrac{d}{dt}(K+U+V) = P_{\text{external}} - \dot{Q}_{\text{outward}} \tag{13.5}$$

$$E_0 = mc^2, \qquad \Delta E_0 = \Delta m\, c^2 \tag{13.6}$$

For a closed-mass body-plus-medium system, K, U and V are kinetic, internal and potential energy. Internal drag transfers K to U; boundary heat and work change the total. Open control volumes use the advective energy flux in (12.3). Equation (13.6) is the rest-energy relation [R10]; a modeled conversion contributes through its explicit source term.

An exact linear-drag example, m $\dot{v} = -c_d v$ in a closed heat bath, gives v_next = $\exp(-c_d \Delta t/m)v$ and heat gain $K - K_{\text{next}}$. The earlier formula-check record gives a direct combined-energy example.

THERMAL / OPTICAL DOMAINS / 16

Diffusion, attenuation and absorption

Thermal and concentration transport

$$\rho c_p(\partial_t \vartheta + \mathbf{v}\cdot\nabla\vartheta) = \nabla\cdot(k\nabla\vartheta) + Q_{\text{heat}} \tag{14.1}$$

$$\partial_t(\rho c) + \nabla\cdot(\rho c\mathbf{v}) = \nabla\cdot(\rho D\nabla c) + S_c \tag{14.2}$$

Equation (14.1) defines a constant-density incompressible thermal adapter with constant specific heat c_p. Compressible transport uses (12.3). In (14.2), c is a dimensionless mass fraction and S_c has units kg/(m³ s). D has units m²/s and k has units W/(m K). Sources include viscous heat, reduced drag or optical absorption.

Amplitude and gyroscopic rotation

$$\dot{A} = -\mu A, \quad A(t) = A_0 e^{-\mu t}, \quad E \propto A^2 \implies E(t) = E_0 e^{-2\mu t} \tag{14.3}$$

$$\mathbf{F}_C = -2m\boldsymbol{\Omega}\times\mathbf{v}, \qquad \mathbf{F}_C\cdot\mathbf{v} = 0 \tag{14.4}$$

Here μ ≥ 0 has units s⁻¹ and denotes attenuation, distinct from viscosity. Amplitude and energy have their respective decay rates. Coriolis acceleration redirects velocity while its instantaneous work is zero. Dissipative channels carry their own power terms.

Optical absorption

$$\frac{dI}{ds} = -\kappa I, \quad I(s) = I(0)e^{-\int_0^s \kappa(\xi)d\xi}, \quad Q_{\text{abs}} = \kappa I \tag{14.5}$$

For a collimated absorbing beam, I is irradiance in W/m² and κ an absorption coefficient in m⁻¹. Q_{abs} is deposited heat per volume. Reflectance, transmittance and absorptance satisfy R+T+A = 1 for the declared optical balance. Scattering adds a radiative-transfer operator [R9].

A dark-material domain supplies wavelength-dependent optical coefficients and measured interface response. Geometry controls paths and surface orientation; material fields determine absorption. The source’s material cue becomes this explicit geometry-material-radiation coupling.

PORT-HAMILTONIAN DOMAIN / 17

Energy-based control and interconnection

A common energy language connects mechanics, circuits, transport and control. Let x be the state and H(x) its stored energy. Define a skew operator J, a symmetric dissipative operator R and an input map G [R12].

$$\dot{x} = (J(x) - R(x))\nabla H(x) + G(x)u, \quad J^T = -J, \quad R = R^T \geqslant 0 \tag{15.1}$$

$$y = G^T \nabla H, \qquad \dot{H} = -\nabla H^T R \nabla H + y^T u \tag{15.2}$$

J redistributes energy, R supplies dissipated power, and $y^T u$ is exchanged input power. The operator domains and units determine the state variables: momentum and position, charge and flux, thermal amounts or cell inventories.

Coupling ports

$$u_1 = C_{\text{port}} y_2, \quad u_2 = -C_{\text{port}}^T y_1, \qquad y_1^T u_1 + y_2^T u_2 = 0 \tag{15.3}$$

C_{port} maps compatible power-conjugate units; normalized ports may use the identity. The transfers are equal and opposite in power. Adding subsystem balances cancels their internal exchange. Shared geometry supplies port areas, normals and adjacency.

An explicit mechanical example

$$H(q,p) = \tfrac{p^2}{2m} + \tfrac{kq^2}{2}, \quad \dot{q} = p/m, \quad \dot{p} = -kq - \gamma p/m + u \tag{15.4}$$

$$\dot{H} = -\gamma (p/m)^2 + (p/m)u \tag{15.5}$$

The oscillator’s dissipated power can enter the thermal adapter. This couples macroscopic motion, heat and control through a common state and energy balance.

The architecture’s general coupling C specifies ports and variable transformations. Port-Hamiltonian structure is the energy-based specialization of that map. Its algebra supplies the balance directly through the update model.

GRAPH / NEURAL DOMAINS / 18

Graph signals, neural fields and connectomes

Interpret the symmetric zero-diagonal mask bits as adjacency Q. Add weighted edges W when interaction strength or direction matters. Entity IDs can name people, cells, neurons, sensors or regions.

$$L = D - Q, \quad D_{ii} = \sum_j Q_{ij}, \qquad \dot{x} = -\kappa L x, \quad \kappa > 0 \tag{16.1}$$

$$x^T L x = \sum_{\{i,j\} \in E} (x_i - x_j)^2, \qquad \mathbf{1}^T L = 0 \tag{16.2}$$

This is graph diffusion. On a connected undirected graph it smooths differences and preserves the total signal. Eigenvectors of L give spectral modes for layout, clustering and multiscale signal analysis.

A neural dynamical specialization

$$\dot{x}_i = -\lambda_i x_i + \sum_j W_{ij} \phi(x_j) + u_i \tag{16.3}$$

λ_i is a local decay rate, φ a selected nonlinear response and u_i an external drive. The hierarchy indexes regions; W stores connectivity, including cycles. This defines the computational-connectome interface through actual signals and interactions.

Message-passing geometry

$$h_i^{\ell+1} = \sigma\left(W_0 h_i^{\ell} + \sum_{j \in \mathcal{N}(i)} a_{ij} W_1 h_j^{\ell} \right) \tag{16.4}$$

A learned graph module transforms features on the same entities. Features can include cell coordinates, material coefficients, primitive parameters or observations. A decoder can map h_i to an implicit leaf or a field coefficient; its training loss becomes part of the fitting channel.

For centered entity coordinates $X \in R^{N\times 3}$ and a chosen graph eigenmode $v \in R^N$, define the spatial axis $a = X^T v$. A nonzero xz projection gives $\beta = \operatorname{atan2}(a_z, a_x)$. Store the selected eigenvalue, normalization, sign and reference frame with this spiral-axis adapter.

Shared operator

One graph can support diffusion, spectral visualization, neural interaction and quantum graph-state description. Each adapter assigns its own state and evolution to the same declared adjacency.

DATA STRUCTURES / COMPUTATION / 19

Packed topology, rank/select and hardware

Logical heap addresses retain the original formula 2^d−1+j. A separate breadth-first shape stream D stores two bits per actual node: left-child presence and right-child presence.

$$D[2i+s] = \mathbf{1}[\text{child slot } s \text{ exists}], \quad s \in \{0,1\} \tag{17.1}$$

$$\text{child}(i,s) = \text{rank}_1(D, 2i+s) \quad \text{when } D[2i+s] = 1 \tag{17.2}$$

$$\text{parent}(j) = \lfloor \text{select}_1(D,j)/2 \rfloor, \quad j \geq 1 \tag{17.3}$$

Rank is inclusive at a zero-based position; select uses a one-based occurrence count and returns a zero-based position. Breadth-first node IDs start at zero. This keeps missing-left and missing-right information. The shape uses 2N bits, with an o(N) auxiliary rank/select index in a succinct implementation [R13].

$$\text{bit}(W,i) = (W_{\lfloor i/64 \rfloor} \gg (i \bmod 64)) \text{AND } 1 \tag{17.4}$$

Payloads, coordinates and parameters have their own packed schemas. SIMD/GPU kernels evaluate independent fields or wordwise masks. Node topology, sign masks and entity records retain separate lengths and offsets.

Electrical and logical timing

$$\tau_{RC} = RC, \qquad D_{\text{path}} = \sum_{g \in \text{path}} d_g \tag{17.5}$$

RC is a circuit time constant in seconds; d_g are propagation delays. These quantities connect a physical implementation to clocked or asynchronous scheduling. The full-turn τ_circle and log-time coordinate remain mathematical controls with their stated domains.

Temporal folding is represented concretely by the dependency graph, reuse schedule and measured path delays. The implementation records word width, address representation, memory traffic and kernel timing.

DISTRIBUTED / CYBERNETIC DOMAINS / 20

Replicated state and self-producing rules

A replica stores a set O of uniquely tagged operations. Each tag contains a replica ID and local sequence number; its payload identifies an entity event, parameter assignment or mask mutation.

$$O_a \sqcup O_b = O_a \cup O_b \tag{18.1}$$

$$A \sqcup B = B \sqcup A, \quad (A \sqcup B) \sqcup C = A \sqcup (B \sqcup C), \quad A \sqcup A = A \tag{18.2}$$

Union is associative, commutative and idempotent. Replicas receiving the same tagged operations therefore obtain the same operation set, independent of delivery order and repetition. This is a state-based join-semilattice construction [R14].

$$b(O) = b_0 \oplus \bigoplus_{o \in O} u(o) \tag{18.3}$$

XOR the payload associated with each distinct operation tag once; two tags carrying the same payload cancel each other. Absolute assignments use a maximum version tuple under a declared total order. Position and organizational-parent assignments retain their respective fields and stable entity IDs.

Self-referential and autopoietic specialization

$$(P_{n+1}, \theta_{n+1}, \Omega_{n+1}) = \mathcal{R}(P_n, \theta_n, \Omega_n, \mathrm{Eval}(P_n, s_n)) \tag{18.4}$$

R is an explicit production rule. It can regenerate its program, parameters and defining boundary from evaluated internal products. A concrete autopoietic module declares the components it produces and the relations by which those components regenerate the module boundary.

Ordinary geometric feedback uses U from page 6. The self-producing specialization adds these regeneration relations. Distributed materialization declares its event ordering for geometry edits and continuous parameter assignments.

> **Formal connection**
>
> Stable IDs support cross-representation identity. The merge algebra supplies replica convergence. The production rule supplies the self-producing cybernetic behavior. These are separately defined operators within shared state.

CODING / QUANTUM INFORMATION / 21

Parity masks and quantum graph states

Retain the double Hadamard product, with elementwise multiplication and sign inputs A,B ∈ {−1,+1}^(N×N).

$$M = (A \odot B) \odot (A^T \odot B^T) = C \odot C^T, \quad C = A \odot B \tag{19.1}$$

$$Q = A_{\text{bit}} + B_{\text{bit}} + A_{\text{bit}}^T + B_{\text{bit}}^T \quad (\mathbb{F}_2) \tag{19.2}$$

The bit/sign convention is $M_{ij} = (-1)^{Q_{ij}}$. M is symmetric with a +1 diagonal. Q is symmetric with a zero diagonal, exactly a simple undirected graph adjacency. The independent mask capacity is N(N−1)/2 bits.

Constructing the stabilizers

$$K_i = X_i \prod_{j\,:\,Q_{ij}=1} Z_j \tag{19.3}$$

$$H = [I \mid Q], \qquad H \Lambda H^T = Q + Q^T = 0 \quad (\mathbb{F}_2) \tag{19.4}$$

Λ is the binary symplectic form exchanging the X and Z halves. Equation (19.4) establishes commutation. The identity block gives N independent generators. Their +1 common eigenspace is the unique graph state for Q [R15].

Coding interpretation

A selected independent commuting subset of r generators defines a stabilizer subspace with N−r logical qubits. The generator phases and chosen subset are explicit quantum-module state. The mask program therefore authors a mathematically defined graph-state description.

Classical parity checks use a binary matrix H_c and syndrome $s = H_c x$ over F_2. Error behavior belongs to the selected code and channel model. The shared architecture stores generators, syndromes, graph relations and their entity/address metadata.

> **Exact bridge**
>
> The original XOR symmetrization creates graph adjacency. Graph adjacency creates commuting Pauli generators. This construction supplies the quantum connection through explicit operators and a defined state space.

BOOLEAN FUNCTIONS / CODING / 22

Nonlinear Boolean and spectral channels

The existing four-term XOR is a linear map over F_2. It supplies fast parity and symmetric graph construction. A separate nonlinear function gives the source's proposed Boolean-spectrum connection an explicit realization.

$$g(x,y) = x \cdot y \pmod 2, \qquad x,y \in \mathbb{F}_2^k \tag{20.1}$$

$$W_g(a,b) = \sum_{x,y} (-1)^{g(x,y) + a\cdot x + b\cdot y} = 2^k (-1)^{a\cdot b} \tag{20.2}$$

The Walsh magnitude is exactly 2^k at every spectral coordinate. This is a bent function on 2k variables [R16]. Its number of ones is 2^(2k−1)−2^(k−1), as follows from $W_g(0,0) = 2^k$.

A four-bit example

$$g(x_1,x_2,x_3,x_4) = x_1x_2 \oplus x_3x_4 \tag{20.3}$$

Across 16 inputs, this function has six ones and all 16 Walsh coefficients have magnitude four. The test record reproduces those values.

Typed composition

A Boolean module stores its function, input partition and output interpretation. Its outputs can select geometry leaves, interaction edges or sampling patterns. Linear parity, nonlinear Boolean features and quantum stabilizers are three separately defined channels that can share packed storage.

A cryptographic construction defines its keyed state transformation. A quantum experiment defines preparation, operations and measurement. These application maps assign operational meaning to the formal functions.

> **Beyond the original mask**
>
> The architecture can carry both the original linear symmetrization and a maximally nonlinear Boolean channel. Their mathematical properties are explicit, so applications can select the operator appropriate to their domain.

RELATIVISTIC DOMAIN / 23

Log-time and 3+1 spacetime geometry

A spacetime adapter uses a three-dimensional spatial manifold Σ, for example a radial shell with angular coordinates (u,ψ). In this page, use geometrized units c = G = 1 [R17].

$$ds^2 = -N^2 dT^2 + h_{ij}(dx^i + \beta^i dT)(dx^j + \beta^j dT) \tag{21.1}$$

h is the spatial metric, N the lapse and β the shift. Their state is supplemented by extrinsic curvature K and matter energy/momentum fields.

$$K_{ij} = -\tfrac{1}{2N}(\partial_T h_{ij} - \mathcal{L}_\beta h_{ij}) \tag{21.2}$$

$$^{(3)}R + K^2 - K_{ij}K^{ij} = 16\pi\rho_E \tag{21.3}$$

$$D_j(K^{ij} - h^{ij}K) = 8\pi j^i \tag{21.4}$$

Here K = $h^{ij}K_ij$ is the trace, $\rho_E = T_{ab}\, n^a n^b$ the normal-frame energy density, and $j^i = -h^i{}_a\, n_b\, T^{ab}$ the projected momentum density. Equations (21.3)-(21.4) are the ADM Hamiltonian and momentum relations with the sign convention (21.2). The completion chapters supply curvature evolution, scalar matter, harmonic gauge, initial data and an exact expanding log-time solution.

The log-time coordinate as an explicit lapse

$$t = t_*(e^T - 1), \qquad N_T = N_t\, t_* e^T, \qquad \beta^i_T = \beta^i_t\, t_* e^T \tag{21.5}$$

Time reparameterization multiplies the lapse and shift by dt/dT. For a static Euclidean shell with zero shift, $h_{rr} = 1$, $h_{uu} = r^2/(1-u^2)$, $h_{\psi\psi} = r^2(1-u^2)$ in its regular chart; K = 0. A unit t-lapse becomes $N_T = t_*e\hat{}T$.

This lifts the source’s log-time idea into a Lorentzian metric representation. Overlapping nonsingular angular charts or Cartesian coordinates cover the poles. Matter, curvature and field evolution occupy their own stored tensor fields.

SPACE / CYBER-PHYSICAL SYSTEMS / 24

Orbital geometry, networks and observation

An orbital entity stores ID, epoch, reference frame, position r and velocity v. The sphere index addresses direction; the radius and velocity remain physical state. Frame and ephemeris transformations follow a declared adapter [R18].

$$\mathbf{s} = \mathbf{r}/\|\mathbf{r}\|, \qquad w = \mathrm{locate}(R\mathbf{s}, d) \tag{22.1}$$

$$\ddot{\mathbf{r}} = -\frac{\mu_g}{\|\mathbf{r}\|^3}\mathbf{r} + \mathbf{a}_{\mathrm{control}} \tag{22.2}$$

For relative two-body position in nonrotating axes, $\mu_g = G(M+m)$, with GM as the test-mass specialization. Perturbations enter as additional acceleration. Coverage queries use spherical cells while the physical trajectory uses (22.2).

Communication geometry

$$\mathbf{y} = H\mathbf{x} + \mathbf{n}, \qquad \mathbf{x} = W\mathbf{s}_{\mathrm{signal}} \tag{22.3}$$

H is a complex channel matrix, W a beamforming map and n the channel noise. Antenna positions, wavelengths, delays and reference-frame transforms determine the chosen channel model. The hierarchy indexes endpoints and coverage regions; the network graph stores communication relations.

Observation and state estimation

$$\dot{x} = f(x,u), \qquad y = h(x) + \nu, \qquad \dot{\hat{x}} = f(\hat{x},u) + K(y - h(\hat{x})) \tag{22.4}$$

The same observation interface applies to robotics, instrumented materials and digital twins. Geometry supplies sensor locations or expected measurements; physical fields supply the evolving state; the observer maps measurements into estimated state.

An observation is timestamped with its frame and units. A controller acts through the corresponding physical port. This gives a concrete cross-layer connection among spatial lookup, field simulation, sensing and network control.

INTEGRATION / 25

Coupled profiles across the domains

A profile selects operators and connects typed outputs to typed inputs. It records its geometry program, material laws, fields, clock mapping, observation functions and evaluation schedule.

Profile	Explicit composition
Thermal geometric twin	CSG boundary → optical path integral → deposited heat Q_{abs} → temperature field → thermal deformation → updated geometry.
Buoyant plume	Volume cells → mass/momentum/energy fluxes → density/velocity/temperature → fitted cap/stem envelope → field rendering and observation.
Graph-neural geometry	Mask Q → graph features → learned residual field → constructive shape → samples → parameter-fitting update.
Replicated hierarchy	Tagged entity operations → set-union merge → deterministic records → organization and spatial lookups → display and aggregate queries.
Quantum graph authoring	Packed A/B inputs → symmetric Q → stabilizer tableau [I|Q] → graph-state preparation description.
Orbital network twin	Ephemeris/frame state → sphere direction/cell → coverage graph → channel model → measurement/control update.

An energy-coupled example

$$\dot{H}_{\mathrm{mech}} = -P_{\mathrm{diss}} + P_{\mathrm{in}}, \qquad \dot{U}_{\mathrm{thermal}} = P_{\mathrm{diss}} - P_{\mathrm{out}} \tag{23.1}$$

$$\frac{d}{dt}(H_{\mathrm{mech}} + U_{\mathrm{thermal}}) = P_{\mathrm{in}} - P_{\mathrm{out}} \tag{23.2}$$

The transfer term cancels when the subsystem balances are added. The same principle governs shared-face fluxes, mechanical/thermal coupling and power-conjugate control ports.

Direct execution

Events and continuous equations update the shared state through U. Empty and full-space geometry retain their defined occupancy values. Each coupling’s equations determine the evolution; source fields, boundary data and model parameters are part of the profile.

The expanded paradigm is therefore a composition of addressable structures, implicit geometry, continuous dynamics and discrete operators. Its domain connections are supplied by the explicit maps in this edition.

CHEMICAL / NON-EQUILIBRIUM DOMAINS / 26

Reaction fields and scale thermodynamics

A reaction-transport adapter uses molar concentrations c = (f,a,b,m)ᵀ in mol/m³, stoichiometric matrix N and rates v(c) in mol/(m³ s). Each reaction column records net species production [R19].

$$\partial_t c + \nabla \cdot (c\mathbf{v}_{\text{flow}}) = Nv(c) + \nabla \cdot (D\nabla c) + s_{\text{in}} - s_{\text{out}} \tag{24.1}$$

$$\ell^T N = 0 \quad \Longrightarrow \quad \ell^T N v(c) = 0 \tag{24.2}$$

A left-null vector l defines a conserved chemical amount. Boundary fluxes and supplied species determine its open-system change. Reaction enthalpies ΔH_r in J/mol contribute the thermal source −Σ_r ΔH_r v_r.

A boundary-producing catalytic example

$$F + B \longrightarrow A + B, \quad F + A \longrightarrow B + A, \quad F + A + B \longrightarrow M + A + B \tag{24.3}$$

$$v_1 = k_1 fb, \quad v_2 = k_2 fa, \quad v_3 = k_3 fab \tag{24.4}$$

F is a precursor, A and B are catalysts, and M is a boundary-forming product. These idealized species share a conserved monomer unit. The stoichiometric columns transfer one F unit to A, B or M, so l = (1,1,1,1) is conserved by the reactions. Rate constants carry the units appropriate to concentration and reaction order.

$$V_M = \overline{v}_M N_s, \qquad R_o = \left(R_i^3 + \tfrac{3V_M}{4\pi}\right)^{1/3} \tag{24.4b}$$

Dissolved product m and deposited shell amount N_s are separate state. Molar volume v̄_M converts N_s into V_M. The completion chapters define deposition, precursor influx, finite reservoir and the global conservation proof [R20].

Scale-patterned transport and entropy

$$k(r) = k_0 e^{\varepsilon \sin(\omega \ln(r/r_*))}, \quad k_0 > 0,\ r_* > 0,\ r > 0 \tag{24.5a}$$

$$k(qr) = k(r), \quad q = e^{2\pi/|\omega|}, \quad \omega \neq 0 \tag{24.5b}$$

$$\dot{S}_{\text{gen, cond}} = \int_{\mathcal{D}} \frac{k\|\nabla\vartheta\|^2}{\vartheta^2}\, dV \geq 0, \quad \vartheta > 0 \tag{24.6}$$

ε and ω are dimensionless; ω = 0 gives constant conductivity. Scale-repeating conductivity gives the log-sine profile a material-transport role. Equation (24.6) is conductive entropy generation; reaction contributions enter separately [R21].

THEOREMS / DEFINITIONS / MODELS / 27

Mathematical status and completion

XD-TAU can be made a coherent cross-domain architecture. Its validity consists of specific algebraic identities, well-defined maps and physical solution theorems. This edition repairs four defects: arbitrary-field CSG, ambiguous seam differentiation, dissolved/shell material double counting, and unsupported discrete conservation. It also supplies the missing evolution and coupling data.

Statement	Mathematical conclusion
Regularized geometry	A complete Boolean algebra of regions; finite arrangement profile is executable exactly.
Coordinates, parity and graphs	Defined maps with their chart, metric and finite-algebra domains.
Finite coupled reference	A causal composition of explicit maps, with energy and amount ledgers.
Membrane reactor	A global, unique nonnegative solution for the specified finite-reservoir ODE.
Fluid continuum	Positive local strong solutions for the specified periodic Newtonian/Fourier family.
Einstein/scalar continuum	Local gauge-fixed development from smooth constraint-satisfying data; an exact expanding log-time profile.
Arbitrary additional coupling	Its own state domain, equations and existence proof must be supplied.

Meaning of complete

A reference profile is complete when its state, operators, coefficients, initial/boundary data and evaluation order determine its evolution. An algebra is complete in the order-theoretic sense when every family has a supremum and infimum. Neither meaning proves global smoothness for every nonlinear continuum system or validates every possible application.

The expansion formalizes connections among domains. For example, a graph adjacency constructs commuting Pauli generators, and an optical path supplies heat to a material model. These are explicit bridges; an analogy alone supplies no additional physical law.

> **Direct execution**
>
> The reference rules use no clipping, rollback, quarantine or rejection stage. Positivity and balances follow from the specified equations and their mathematical domains. The domains are printed with the theorems they determine.

GEOMETRY / PROOF / 28

Regular-closed algebra and exact solids

Let RO(X) contain the regular-open sets U = int(cl U). Interior and closure give inverse bijections RC(X) ↔ RO(X). On RO(X), define:

$$U \wedge V = U \cap V, \quad U \vee V = \operatorname{int}\operatorname{cl}(U \cup V), \quad \neg U = X \setminus \operatorname{cl} U \tag{A.1}$$

Meet and join have their order meanings and complement satisfies U ∧ ¬U = ∅, U ∨ ¬U = X. For distributivity, V ∪ W is dense in V ∨ W. Intersecting that dense set with open U preserves relative density. Taking interior of closure proves U ∩ (V ∨ W) = (U ∩ V) ∨ (U ∩ W). Transporting these laws by closure proves the region operations on page 4 form a Boolean algebra [R22].

$$A \wedge A^{*} = \varnothing, \quad A \vee A^{*} = X, \quad A \setminus_{*} A = \varnothing \tag{A.2}$$

$$\bigvee_i A_i = \operatorname{cl}\left(\bigcup_i A_i\right), \qquad \bigwedge_i A_i = \rho\left(\bigcap_i A_i\right) \tag{A.3}$$

The union of interiors is dense in the union of regular-closed sets; its closure is regular closed. Any regular-closed lower bound of all A_i lies inside ρ(∩A_i). These facts prove (A.3), including infinite families.

Finite exact realization

$$A_I = \operatorname{cl}\left(\bigcup_{j \in I} C_j\right), \quad I \subseteq \{1, \ldots, M\} \tag{A.4}$$

Use a finite polyhedral arrangement of disjoint full-dimensional open cells C_j. Their closures cover X; the cell skeleton has empty interior. A region is the closure of its selected cells. Union, meet and complement correspond exactly to index union, intersection and complement. Contact faces, curves and points have their own feature records.

Refine two arrangements to their common arrangement before combining them. Rational plane data permit exact predicates and constructions. For arbitrary black-box fields, I gives denotational meaning; a finite sample cannot determine all topology between samples.

Zero-plateau repair

f ≡ −1 imports X, while g ≡ 0 imports ∅. Raw max(f,−g) ≡ 0 incorrectly imports ∅. Region-first subtraction returns X, with canonical field −ell. One final regularization of arbitrary raw expressions is not a Boolean homomorphism.

METRIC GEOMETRY / PROOF / 29

Canonical fields and controlled blending

For a nonempty proper regular-closed A, d_A is negative on int A, zero on ∂A and positive outside A. On one side of the boundary its 1-Lipschitz bound is the distance inequality. Across sides, a segment crosses ∂A; the two distances to that crossing sum to at most its length. Thus |d_A(x)−d_A(y)| ≤ |x−y|.

$$\left|\frac{d}{dd}\frac{\ell d}{\ell+|d|}\right| = \frac{\ell^2}{(\ell+|d|)^2} \le 1 \tag{A.5}$$

The finite canonical field inherits the bound and preserves all signs. Since A = cl(int A), closure of its negative set recovers A. Negation produces the complementary interior before closure. Min and max produce the union and common interior; this proves the canonical field identities on page 4.

Smooth-node definition

$$U_h(A,B) = \rho\{x : S_h(d_A(x), d_B(x)) < 0\} \tag{A.6}$$

Use T_h for smooth intersection and T_h(d_A,−d_B) for smooth difference. At tagged empty/full inputs evaluate S_h = min and T_h = max by their limits. No infinite subtraction is formed. For finite a,b, rationalizing the square root gives:

$$\min(a,b) - S_h(a,b) = \frac{h^2}{\sqrt{(a-b)^2+2h^2}+|a-b|} \le \frac{h}{\sqrt{2}} \tag{A.7}$$

$$A \cup B \subseteq U_h(A,B) \subseteq \{x : \operatorname{dist}(x, A\cup B) \le h/\sqrt{2}\} \tag{A.8}$$

For the first inclusion, the negative interior of either input is included before closure. Outside their union, min(d_A,d_B) is precisely the distance to that union; (A.7) gives the second inclusion. Smooth intersections and differences are inner deformations. Thin components can disappear, so a two-sided Hausdorff estimate requires additional thickness assumptions.

Degenerate inputs remain defined

The field x_1^2 imports ∅; $-x_1^2$ imports X. Opposing closed half-spaces have regularized meet ∅ and union X, while their shared plane remains a feature. A supplied region is authoritative even if an auxiliary fitting field is zero.

S_h(d,d) = d−h/√2. Smooth blending therefore changes a shape even for equal inputs. Its derivatives are valid smooth scalar derivatives; idempotence and Boolean laws belong to the exact region algebra.

NONSMOOTH GEOMETRY / FORMAL RULE / 30

Derivatives through seams and masks

For a locally Lipschitz final scalar F on an open Euclidean domain, define its Clarke generalized gradient from attainable gradients of the whole expression [R23]:

$$\partial^C F(x) = \mathrm{co}\,\{\lim_k \nabla F(x_k) : x_k \to x,\ F \text{ differentiable at } x_k\} \tag{A.9}$$

$$g^\circ(x) = \mathrm{argmin}_{g \in \partial^C F(x)} g^T G^{-1} g, \qquad G \succ 0 \tag{A.10}$$

The generalized gradient is nonempty, compact and convex. Strict convexity gives a unique minimum-norm selection. Where F is C¹ locally, it equals ∇F. Differentiability at one point alone does not imply this equality: F(x)=x+x²sin(1/x), F(0)=0, has F′(0)=1 but ∂^CF(0)=[0,2].

Differentiate the final expression

$$F(x) = \max(\min(x, -x), 0) \equiv 0, \qquad \partial^C F(0) = \{0\} \tag{A.11}$$

Recursive branch tie choices can incorrectly return 1 in (A.11). For a single min/max of C¹ branches, the active-gradient convex hull is correct. For nested expressions it can be only an outer bound. Use attainable gradients of the final expression or an exact simplification that preserves F.

Discontinuous sign masks

Let m=m_j on a finite Lipschitz cell partition and let f be C¹. Inside cells, ∇(mf)=m_j∇f. Across seam S, orient n_S from the minus cell to the plus cell. The distributional derivative is:

$$D(mf) = \sum_j m_j \nabla f \mathbb{1}_{C_j}\, dx + \sum_S (m_+ - m_-) f n_S \mathcal{H}^2|_S \tag{A.12}$$

Integration by parts on each cell proves (A.12); opposing traces leave the jump term. A discontinuous field has no Clarke gradient at its jump. Its imported region still has a Lipschitz canonical field, whose spatial generalized derivative is defined.

Spatial Lipschitzness does not imply parameter continuity through component disappearance. Parameter fitting uses its actual finite objective on a locally parameter-Lipschitz domain. Mask and topology edits are discrete transitions. Normals require a regular level-set point with $\nabla\Phi \neq 0$.

STATE / EVENTS / REPRODUCTION / 31

Semantic state and finite encoding

The semantic state contains real fields and parameters, regular-closed regions, finite programs, entity relations, physical inventories and event sets. The finite operational state contains explicit arrays and finite descriptors. Distinguish these two state spaces.

$$\mathrm{Decode}(\mathrm{Encode}(Q_\delta s)) = Q_\delta s \tag{B.1}$$

No injection of arbitrary R^N into finite bit strings exists: the former is uncountable and the latter countable. Q_δ is the recorded quantizer. For rational state, store each reduced numerator/positive denominator exactly. For IEEE-754 arrays, store format, bytes and endian order; this recovers those machine values, rather than every real value.

Deterministic events

Channel	Materialization rule
Operation set	Union uniquely tagged (replica ID, sequence) records.
XOR bits	Apply each distinct tagged operation once; equal payloads under different tags apply separately.
Absolute assignments	Choose maximum (logical time, replica ID, sequence) tuple per field.
Program edits	Sort the same tuple order; apply a total syntax-rewrite function. A missing named subtree leaves the expression unchanged.
Organization graph	Store parent relations as a functional graph. Display cycles as strongly connected groups ordered by stable ID.
Spatial tree	Use the finite binary shape stream and explicit child slots; entity membership is separate from organizational edges.

Total syntax rewrites are defined functions, including their missing-target result. Union is associative, commutative and idempotent; equal operation sets materialize equally because the order and all tie conventions are fixed.

Recorded numerical identity

A snapshot includes step index, physical/log time, ordered schema, coefficients, quantizer, geometry program and numerical scheme. A stochastic adapter additionally stores its generator algorithm, state and draw order. Reference calculations below distinguish exact rational balances from IEEE residuals.

The authoritative geometry representation is either an exact cell arrangement or a declared finite program with its semantic interpretation. Sampled occupancy is an approximation with a recorded resolution, not an exact encoding of an arbitrary continuum boundary.

EXPLICIT COMPOSITION / COUPLINGS / 32

A causal cross-domain update

$$\mathcal{U}_h = O \circ N \circ T \circ A \circ C \circ V \circ F \circ D \circ M \circ E \tag{B.2}$$

Read right to left: materialize events E; mechanical midpoint M; body/medium drag D; Euler face fluxes F; viscous conversion V; chemical substeps C; optical absorption A; thermal/species transport T; neural update N; geometry and observation outputs O. Each stage reads the preceding stage. Different profiles record a different explicit composition.

At step start, construct adjacency, source schedules, face data and ray intersections from the materialized state. Freeze each stage's coefficients during that stage. The reference mechanical/drag stages use separate body, medium and heat states, so they do not change the fluid state used for the Euler wave speeds. Updated outputs affect the next scheduled coefficient evaluation.

$$\vartheta_i = E_i/C_i, \quad \kappa_i = \kappa_0 e^{\beta(\vartheta_i - \vartheta_0)}, \quad g_{ij} = g_0 Q_{ij} e^{-|X_i - X_j|^2/\ell_g^2} \tag{B.3}$$

$$R_i = R_{i0} e^{\alpha_i(\vartheta_i - \vartheta_0)}, \qquad r_j = r_{j0} e^{(D_h h + d)_j} \tag{B.4}$$

C_i > 0 is heat capacity; κ₀,g₀ ≥ 0; ell_g > 0; α_i and β have inverse-temperature units. Q is the symmetric graph from the parity mask. X_i are mechanical positions. D_h is a fixed decoder matrix and h the neural feature vector. The positive radii translate/scale declared primitives; rendering then derives occupancy and samples.

Equations (B.3)-(B.6) define one formal coupling family. The executable rational and periodic-fluid specializations use the distinct concrete coefficient choices listed in their reference-data chapter.

$$W = (w_0/N)Q, \quad \dot{h} = -\Lambda h + W\tanh h + u, \quad \Lambda = \mathrm{diag}(\lambda_i),\ \lambda_i \geq 0 \tag{B.5}$$

$$h^+ = e^{-h_t\Lambda} h + h_t \varphi_1(-h_t\Lambda)(W\tanh h + u), \quad \varphi_1(z) = (e^z - 1)/z \tag{B.6}$$

Here h_t is the time step and φ₁(0)=1. Tanh is globally 1-Lipschitz, so the finite neural ODE is globally Lipschitz and has a unique global solution for bounded continuous inputs. (B.6) is its explicit exponential-Euler approximation.

Each stage has the maps and domains stated below. Their composition is consequently single valued on the profile domain. Internal power and amount transfers use one shared ledger entry, so they cancel exactly when balances are added.

MECHANICS / CIRCUITS / CONTROL / 33

A unique discrete energy update

Use a fixed finite quadratic energy $H(x)=\frac{1}{2}x^{T}Qx$, $Q=Q^{T} \geq 0$. Let $J^{T}=-J$, $R=R^{T} \geq 0$, and B map the supplied input u. All are fixed during the substep. Skew port interconnections can be included in J [R24].

$$A = (J - R)Q, \quad M_h = I - \tfrac{h}{2}A \tag{C.1}$$

$$x^{+} = M_h^{-1}\left[\left(I + \tfrac{h}{2}A\right)x + hBu\right], \qquad h \geq 0 \tag{C.2}$$

Existence and uniqueness for every step

If $M_h v=0$, then $v=(h/2)(J-R)Qv$. Multiplying by $v^{T}Q$ gives $v^{T}Qv=-(h/2)(Qv)^{T}R(Qv)\leq 0$. Since Q is positive semidefinite, $Qv=0$, and the original equation gives $v=0$. Thus M_h is invertible, including semidefinite energy and $h=0$.

$$g = Q(x + x^{+})/2, \quad y = B^{T}g, \qquad H(x^{+}) - H(x) = -hg^{T}Rg + hy^{T}u \tag{C.3}$$

Proof: the quadratic difference is $g^{T}(x^{+}-x)$. Substitute $x^{+}-x=h[(J-R)g+Bu]$; skew symmetry removes $g^{T}Jg$. This is an exact discrete identity, rather than a continuous identity applied to an arbitrary numerical step.

$$E_{\text{heat}}^{+} = E_{\text{heat}} + hg^{T}Rg, \qquad \Delta(H + E_{\text{heat}}) = hy^{T}u \tag{C.4}$$

Block Q, J and R represent elastic modes, oscillators, charge/flux circuits and gyroscopic control. Constant matrices give second-order midpoint accuracy. Freezing state-dependent coefficients at stage start retains the balance but generally gives first-order consistency.

Geometry-dependent stiffness

$$W_{\text{parameter}} = \tfrac{1}{2}x^{T}(Q_{\text{new}} - Q_{\text{old}})x \tag{C.5}$$

If a geometry event changes Q, record (C.5) before the mechanical substep. It is parameter work, not unexplained energy creation. The source ledger includes it along with external port work.

> **Failure regression**
>
> For the unit oscillator, forward Euler gives $H^{+}=(1+h^{2})H$. A sound continuous energy law therefore does not prove conservation by an arbitrary discrete update. Equation (C.2) supplies the missing discrete proof.

THERMAL / SPECIES / GRAPH PROOF / 34

Positive transport and heat exchange

Let n_i be inventories in fixed cells of volume V_i > 0; c_i=n_i/V_i. Let g_ij=g_ji ≥ 0 and L_g be the conductance Laplacian. Define:

$$\dot{n} = -L_g V^{-1} n = An, \qquad n^+ = e^{hA} n \tag{C.6}$$

A has nonnegative off-diagonal entries and $1^T A=0$. Choose λ large enough that A+λI is entrywise nonnegative. The series e^(hA)=e^(−λh)Σ_k h^k(A+λI)^k/k! is entrywise nonnegative. Also $1^T e^{(hA)}=1^T$. Hence (C.6) preserves nonnegativity and total inventory for every h ≥ 0.

Two-cell physical map

$$\bar{c} = \frac{V_i c_i + V_j c_j}{V_i + V_j}, \quad r = e^{-gh(1/V_i + 1/V_j)}, \quad \delta = c_i - c_j \tag{C.7}$$

$$c_i^+ = \bar{c} + \frac{V_j r \delta}{V_i + V_j}, \qquad c_j^+ = \bar{c} - \frac{V_i r \delta}{V_i + V_j} \tag{C.8}$$

Both outputs are convex combinations of the inputs, and V_ic_i⁺+V_jc_j⁺ is unchanged. Iterate edges in a recorded order. A palindromic half-step composition is the usual second-order splitting for fixed conductances. A single ordered sweep is first order.

For heat, replace V_i by heat capacity C_i and c_i by absolute temperature θ_i. Energy is Σ_i C_iθ_i. For θ_i > 0:

$$\frac{d}{dt}\sum_i C_i \ln \vartheta_i = \sum_{\{i,j\}} g_{ij} \frac{(\vartheta_i - \vartheta_j)^2}{\vartheta_i \vartheta_j} \geq 0 \tag{C.9}$$

The exact pair map preserves the same entropy sign. The rational alternative r=[1+gh(1/V_i+1/V_j)]⁻¹ is backward Euler for this pair difference; it preserves positivity, amount and the entropy sign. It is a first-order approximation, not the exact exponential flow.

The rational reference uses this alternative so every finite operation and ledger can be verified exactly with reduced fractions. General exponential maps use real semantics and separately reported floating-point residuals.

CHEMICAL DYNAMICS / PROOF / 35

Reaction subflows and chemical energy

In a fixed cell use nonnegative concentrations (f,a,b,m). For each catalytic reaction from page 26, hold its unchanged catalyst fixed during that subflow. The maps for a time h ≥ 0 are:

$$R_1:\ f^+ = fe^{-k_1 bh},\quad a^+ = a + f - f^+,\quad b^+ = b,\quad m^+ = m \tag{C.10}$$

$$R_2:\ f^+ = fe^{-k_2 ah},\quad b^+ = b + f - f^+,\quad a^+ = a,\quad m^+ = m \tag{C.11}$$

$$R_3:\ f^+ = fe^{-k_3 abh},\quad m^+ = m + f - f^+,\quad a^+ = a,\quad b^+ = b \tag{C.12}$$

Each consumed amount $f - f^+$ is nonnegative and is added exactly once to its product. Thus every map preserves the nonnegative cone and f+a+b+m. $R_3 \circ R_2 \circ R_1$ is an explicit first-order Lie splitting; a symmetric half-step composition gives second order. It approximates the simultaneous ODE, rather than equaling its flow.

Global continuous reaction solution

The simultaneous polynomial mass-action vector field is locally Lipschitz. At each coordinate face its derivative is inward or tangent. Conservation of f+a+b+m bounds every nonnegative coordinate. Local solutions therefore remain in a compact invariant set and extend uniquely for all forward time.

$$E_{\mathrm{chem}} = V e^T c, \qquad \Delta E_{\mathrm{heat}} = -V e^T (c^+ - c) \tag{C.13}$$

Here V is cell volume and e contains prescribed species energies in J/mol. The transfer in (C.13) makes chemical-plus-thermal energy cancel exactly. The exothermic reference chooses e_F ≥ e_A,e_B,e_M, so reaction heat is nonnegative and positive initial thermal energy remains positive.

Deposition into a shell

$$m^+ = me^{-k_d h}, \qquad N_s^+ = N_s + V(m - m^+) \tag{C.14}$$

For a shell/reservoir profile, use total chemical energy $V\, e^T c + e_s N_s + V_b\, e_F\, f_b$. Deposit heat $(e_M - e_s)V(m - m^+)$; $e_M \geq e_s$ gives exothermic positivity. Equal precursor energies make reservoir exchange cancel. These stages give a positive, amount-conserving split approximation to the next page's ODE.

The finite rational profile replaces $\exp(-kh)$ by $(1+kh)^{-1}$ at each subflow. These backward-Euler subflows preserve the same transfers and signs and are first-order consistent. Their exact fraction balances do not turn them into exact exponential solutions.

PHYSICAL FEEDBACK / GLOBAL THEOREM / 36

A closed self-producing membrane

Take fixed core radius R_i > 0, volume V=4πR_i³/3 and reservoir volume V_b > 0. State is (f,a,b,m,f_b,N_s), with nonnegative concentrations and N_s(0) > 0. Set k₁,k₂,k₃,k_d,D_s,K_part ≥ 0 and molar volume v̄_M > 0.

$$R_o = \left(R_i^3 + \frac{3\bar{v}_M N_s}{4\pi}\right)^{1/3}, \quad G = 4\pi D_s K_{\text{part}} \frac{R_i R_o}{R_o - R_i} \tag{D.1}$$

$$J = G(f_b - f), \quad v_1 = k_1 fb, \quad v_2 = k_2 fa, \quad v_3 = k_3 fab, \quad v_d = k_d m \tag{D.2}$$

In v₁, fb means f·b; f_b names reservoir concentration. Units: k₁,k₂ m³/(mol s); k₃ m⁶/(mol² s); k_d s⁻¹; D_s m²/s; K_part dimensionless; v̄_M m³/mol; G m³/s; J mol/s. Steady radial Fick diffusion through the shell gives (D.1).

$$V\dot{f} = J - V(v_1 + v_2 + v_3), \quad \dot{a} = v_1, \quad \dot{b} = v_2 \tag{D.3}$$

$$\dot{m} = v_3 - v_d, \quad \dot{N}_s = V v_d, \quad V_b \dot{f}_b = -J \tag{D.4}$$

$$\mathcal{M} = V(f + a + b + m) + N_s + V_b f_b, \qquad \dot{\mathcal{M}} = 0 \tag{D.5}$$

Proof of global existence and nonnegativity

On f=0, ḟ=Gf_b/V ≥ 0; on a=0 and b=0 their derivatives are nonnegative; on m=0, ṁ=v₃ ≥ 0; on f_b=0, ḟ_b=Gf/V_b ≥ 0. Also N_s ≥ 0, so N_s ≥ N_s(0). Adding (D.3)-(D.4) proves (D.5). Every concentration and N_s is bounded by this conserved amount and the positive volumes.

The initially positive shell keeps R_o−R_i > 0. G is smooth and bounded on the compact invariant state set. The vector field is locally Lipschitz on a neighborhood of that set. Picard–Lindelöf and continuation yield a unique global nonnegative solution for all t ≥ 0.

Actual regenerative connection

Internal reactions produce dissolved product; deposition creates the shell; shell thickness changes precursor conductance; precursor flux changes reactions. Positive initial a,b seed catalysis. Available precursor and positive relevant rates produce material. This is a specified self-producing boundary model with a closed material account.

NONLINEAR FLUID DISCRETIZATION / 37

Compressible flow: an actual face map

Use integrated conservation on a fixed periodic polyhedral mesh. For each cell, U=(ρ,m,ℰ) contains density, momentum density and total-energy density. Let γ=1+R_s/c_v > 1 and define:

$$\mathcal{G} = \{U : \rho > 0,\ \mathcal{E} - |m|^2/(2\rho) > 0\}, \quad p = (\gamma - 1)(\mathcal{E} - |m|^2/(2\rho)) \tag{E.1}$$

$$F_n(U) = (m \cdot n,\ (m \otimes m/\rho + pI)n,\ (\mathcal{E} + p)(m \cdot n)/\rho) \tag{E.2}$$

$$\widehat{F}_{ij} = \frac{F_{n_{ij}}(U_i) + F_{n_{ij}}(U_j) - \alpha_{ij}(U_j - U_i)}{2} \tag{E.3}$$

$$\alpha_{ij} = \max_{k \in \{i,j\}} \left(|v_k \cdot n_{ij}| + \sqrt{\gamma p_k/\rho_k} \right) \tag{E.4}$$

$$U_i^+ = U_i - \frac{h}{V_i} \sum_{f = ij} A_f \widehat{F}_{ij}, \quad h = \sigma \min_i \frac{2V_i}{\sum_{f = ij} A_f \alpha_{ij}}, \quad 0 < \sigma \leq 1 \tag{E.5}$$

Normals point outward and ∑_f A_fn_f=0. The same face flux is used with opposite sign in its neighbor; summing (E.5) conserves total mass, momentum and energy exactly in real arithmetic. The displayed h directly defines the step from the snapshot wave speeds.

Positivity proof

$$U_i^+ = (1 - \sum_f w_f) U_i + \sum_{f = ij} w_f (U_j - F_{n_{ij}}(U_j)/\alpha_{ij}), \quad w_f = \frac{h A_f \alpha_{ij}}{2V_i} \tag{E.6}$$

The set G is convex because |m|²/(2ρ) is convex for ρ > 0. For a split state U±F_n(U)/α, put a=1±v_n/α. Its density is aρ and its internal energy density is aρe−p²/(2ρaα²). Since aα ≥ c=√(γp/ρ), that expression is strictly positive for γ > 1. Equation (E.5) makes all weights nonnegative with sum at most one, proving U_i⁺ ∈ G [R25].

This is a concrete first-order Rusanov Euler stage. The positive-temperature Newtonian/Fourier continuum and the compatible viscous stage are specified separately. A flux definition alone does not prove convergence of every coupled mesh/geometry scheme.

COMPATIBLE DISCRETE ACCOUNTING / 38

Viscous heat and moving volumes

After Euler, keep cell masses M_i=V_iρ_i fixed during viscosity. Let p collect integrated momenta, M be the repeated mass diagonal, and B the discrete strain operator. Set W_e=w_eI, w_e ≥ 0, and C_vis,e=C_vis,e^{T} ≥ 0. Their product is symmetric positive semidefinite; D=B^{T}W C_visB ≥ 0.

$$p^{+} = (I + \tfrac{h}{2}DM^{-1})^{-1}(I - \tfrac{h}{2}DM^{-1})p, \quad \bar{v} = M^{-1}(p + p^{+})/2 \tag{E.7}$$

$$\Delta K = -h\bar{v}^{T}D\bar{v}, \qquad K = \tfrac{1}{2}p^{T}M^{-1}p \tag{E.8}$$

The proof (C.1)-(C.4) applies with J=0, Q=M^{-1}. If B annihilates constant translations, total momentum is preserved. Distribute element dissipation with nonnegative shares a_ie, ∑_i a_ie=1:

$$q_i = h\sum_{e} a_{ie}(B\bar{v})_e^{T}W_eC_{\mathrm{vis},e}(B\bar{v})_e \geq 0 \tag{E.9}$$

$$(V_i\mathcal{E}_i)^{+} = (V_i\mathcal{E}_i) + (K_i^{+} - K_i) + q_i \tag{E.10}$$

The internal energy in cell i increases by q_i. Summing (E.10) cancels kinetic loss against heat gain, preserving total energy. Follow with the positive heat-exchange stage using cell heat capacity M_ic_v. This gives an explicit compatible viscous approximation; choose B and element weights as part of the mesh data.

A moving geometry is a different control-volume rule

$$\frac{d}{dt}\int_{K(t)} U\,dV = -\int_{\partial K(t)} (F(U) - Uw)\cdot n\,dA + \int_{K(t)} S\,dV \tag{E.11}$$

Here w is mesh-boundary velocity. Reynolds transport gives (E.11). A discrete ALE scheme uses one swept face volume ΔV_f for both cell-volume change and mesh-advection flux, with opposite neighbor signs:

$$V_i^{+} = V_i + \sum_f \Delta V_{if}, \qquad (VU)_i^{+} = (VU)_i - h\sum_f A_f\hat{F}_f + \sum_f U_f\Delta V_{if} \tag{E.12}$$

For constant U and zero physical flux divergence, the same ΔV_f gives (VU)_i^{+}=U V_i^{+}; this is the geometric conservation identity. The executable reference uses a fixed mesh, so ΔV_f=0. Rendered CSG changes do not silently change its physical cell volumes. A moving-boundary application must supply its swept geometry, boundary traction and a corresponding positivity theorem.

MECHANICAL / THERMAL / OPTICAL PROOF / 39

Drag, optics and a combined ledger

For body and medium masses $m_1, m_2 > 0$, use internal forces $\pm c|w|w$, $c \geq 0$. Put $w = v_1 - v_2$, $M = m_1 + m_2$, $\mu = m_1 m_2 / M$ and $V = (m_1 v_1 + m_2 v_2)/M$. The exact relative-drag flow is:

$$w^+ = \frac{w}{1 + (c/\mu)|w|h}, \quad v_1^+ = V + (m_2/M)w^+, \quad v_2^+ = V - (m_1/M)w^+ \tag{F.1}$$

$$\Delta E_{\text{heat}} = \frac{\mu}{2}(|w|^2 - |w^+|^2) \geq 0 \tag{F.2}$$

Momentum is unchanged. $K = \tfrac{1}{2}M|V|^2 + \tfrac{1}{2}\mu|w|^2$ proves exact kinetic-plus-thermal conservation. At $w = 0$ the displayed map directly returns zero. Linear drag replaces the reciprocal factor by $\exp(-\gamma h/\mu)$.

Rotation and attenuation

$$v^+ = \exp(-2h[\Omega]_\times)v, \qquad (v^+)^T v^+ = v^T v \tag{F.3}$$

The cross-product matrix is skew, so its exponential is orthogonal. This is the finite Coriolis rotation, with kinetic energy unchanged. For ordered constant-absorption ray segments:

$$E_{r,j+1} = E_{r,j} e^{-\kappa_j \ell_j}, \qquad \Delta E_{\text{heat},j} = E_{r,j} - E_{r,j+1} \tag{F.4}$$

$\kappa_j \geq 0$ and length $\ell_j \geq 0$ imply nonnegative transmitted energy and heat for nonnegative incidence. One packet moves through segments in order. Incident packet energy is either existing optical state or an external source entry; it is never counted twice.

Composition theorem

$$\Delta \mathcal{E}_{\text{total}} = W_{\text{ports}} + W_{\text{parameter}} + E_{\text{incoming}} - E_{\text{outgoing}} \tag{F.5}$$

Count mechanical quadratic storage including body kinetic energy, separate medium kinetic energy, fluid total energy, thermal reservoirs, chemical and shell species energy, and remaining optical packets. Count every degree of freedom once. The preceding stage identities telescope; internal transfers cancel. This proves (F.5) for their causal composition in exact arithmetic.

If a physical degree of freedom belongs to the fluid, its viscous heat is already inside fluid energy. A reduced drag reservoir may couple to that fluid only with its equal/opposite momentum and shared heat transfer recorded. The reference uses disjoint subsystem states and a common source ledger.

CONTINUUM / OBSERVATION / NETWORKS / 40

Initial data and existence domains

For the compressible Newtonian/Fourier continuum, choose the periodic domain T^3, R_s,c_v,μ,$k > 0$ and $\lambda_v \geq 0$. Set b=Q=0 for the closed reference. Smooth initial density and temperature are strictly positive; take periodic H^s data, $s \geq 6$. These data determine a unique positive local strong solution [R26].

$$s = c_v \ln(\vartheta/\vartheta_{\mathrm{ref}}) - R_s \ln(\rho/\rho_{\mathrm{ref}}) \tag{G.1}$$

$$\partial_t(\rho s) + \nabla \cdot (\rho s v + q/\vartheta) = \frac{\tau : \nabla v}{\vartheta} + \frac{k|\nabla\vartheta|^2}{\vartheta^2} + \frac{Q}{\vartheta} \tag{G.2}$$

$$\tau : \nabla v = 2\mu |D_{\mathrm{dev}}|^2 + (\lambda_v + 2\mu/3)(\nabla \cdot v)^2 \geq 0 \tag{G.3}$$

The reference periodic problem has no external boundary. An inlet, wall, traction or thermal-boundary problem is a separate data set with its boundary flux account. Global smoothness for arbitrary large three-dimensional fluid data is not established by this local theorem.

Explicit sensing and communication specialization

$$\dot{x} = A_c x + B_c u, \quad y = C_c x, \quad \dot{\hat{x}} = A_c \hat{x} + B_c u + K(y - C_c \hat{x}) \tag{G.4}$$

Fix finite matrices A_c,B_c,C_c,K and a continuous input. This linear observer has a unique global solution. Its error satisfies ė=(A_c−KC_c)e; convergence requires that matrix to be Hurwitz. The architecture does not infer convergence from the observer syntax alone.

$$H_{ij} = a_{ij} \frac{e^{-2\pi i r_{ij}/\lambda}}{r_{ij}}, \quad r_{ij} = |X_i - X_j| > 0, \quad \lambda > 0 \tag{G.5}$$

This scalar free-space channel has stored complex gains a_ij and wavelength. Set W=I and noise zero for its deterministic reference. Detailed antenna/scattering models define their own coefficients. Orbital evolution on page 24 is locally unique for $r \neq 0$ with continuous prescribed acceleration, or locally Lipschitz state-dependent controls, until collision or the end of those data.

Finite coupled ODEs with locally Lipschitz constitutive maps have local unique solutions. On a ball inside the state domain with vector-field bound M and Lipschitz constant L, hM ≤ r and hL < 1 give the contraction proof. Global continuation needs a compact invariant subset inside that domain or another continuation bound; the membrane model supplies one.

SPACETIME / COMPLETE EQUATIONS / 41

ADM evolution with scalar matter

Use signature (−+++), c=G=1, K_{ij}=−½L_nh_ij and ∂⊥=∂_T−L_β. The metric h is positive definite, lapse N > 0 and K symmetric. D is the Levi-Civita derivative of h; R_ij its spatial Ricci tensor [R27].

$$\partial_\perp h_{ij} = -2NK_{ij} \tag{H.1}$$

$$A_{ij} = -D_iD_jN + N(R_{ij} + KK_{ij} - 2K_{ik}K^k{}_j) \tag{H.2a}$$

$$\partial_\perp K_{ij} = A_{ij} - 8\pi N[S_{ij} - \tfrac{1}{2}h_{ij}(S - \rho_E)] \tag{H.2b}$$

Here S=h^{ij}S_ij. Keep the Hamiltonian/momentum constraints (21.3)-(21.4). A smooth potential V(φ) specifies scalar matter; the reference uses V=0.

$$T_{ab} = \nabla_a\phi\nabla_b\phi - g_{ab}(\tfrac{1}{2}\nabla_c\phi\nabla^c\phi + V), \quad \nabla^a\nabla_a\phi - V' = 0 \tag{H.3}$$

$$\Pi = n^a\nabla_a\phi, \quad \rho_E = \tfrac{1}{2}(\Pi^2 + |D\phi|^2) + V, \quad j_i = -\Pi D_i\phi \tag{H.4}$$

$$S_{ij} = D_i\phi D_j\phi + h_{ij}(\tfrac{1}{2}\Pi^2 - \tfrac{1}{2}|D\phi|^2 - V) \tag{H.5}$$

$$(\partial_T - \beta^i\partial_i)\phi = N\Pi \tag{H.6}$$

$$(\partial_T - \beta^i\partial_i)\Pi = N(D^iD_i\phi + K\Pi - V') + D^iND_i\phi \tag{H.7}$$

These equations determine every matter contribution instead of treating energy density, momentum and stress as unrelated arrays. They also fix the sign conventions needed to compare ADM expressions.

This continuum adapter represents gravitational and scalar-field state. Its coordinate and entity links organize that state. The existence theorem on the next page uses a specified hyperbolic gauge; the algebraic constraints alone do not determine an evolution.

RELATIVISTIC CAUCHY PROFILE / 42

Harmonic gauge and constraint propagation

Choose a periodic Cartesian atlas on T^3 and harmonic spacetime coordinates. With $\Gamma^i = h^{jk}\Gamma^i{}_{jk}$, the lapse and shift obey:

$$(\partial_T - \beta^j \partial_j) N = -N^2 K \tag{H.8}$$

$$(\partial_T - \beta^j \partial_j) \beta^i = N^2 \Gamma^i - N D^i N \tag{H.9}$$

Equivalently, define the contracted spacetime connection $C^a = g^{bc}\Gamma^a{}_{bc}$. The reduced Einstein/scalar equation is:

$$R_{ab} - \nabla_{(a} C_{b)} = 8\pi(\nabla_a \phi \nabla_b \phi + V g_{ab}) \tag{H.10}$$

For a general atlas use a harmonic-map gauge with $\Gamma - \Gamma$ relative to a fixed background connection. The periodic reference has no spatial boundary. Specify smooth initial (h, K, φ, Π) satisfying the four ADM constraints, smooth positive N and smooth β. The metric, matter and gauge equations determine initial time derivatives.

Local well-posedness

The reduced equations have a quasilinear wave principal part. Smooth constraint-satisfying Cauchy data yield a unique local gauge-fixed development; the corresponding spacetime is unique up to the usual diffeomorphism equivalence. The Bianchi identity and matter equation give a homogeneous subsidiary wave equation for C. Vanishing initial gauge/constraint data therefore propagate [R28].

This theorem applies on the local smooth solution interval. It does not assert global smooth evolution for every gravitational initial state. The next page gives an explicit nontrivial solution defined for all forward log times in its specified family.

Physical domain ledger

Expression	Required domain
Material mechanics	Positive masses/densities; $\mu_{el} > 0$ and $\lambda_{el} + 2\mu_{el}/3 > 0$; $\det D\chi > 0$ for finite deformation.
Heat and diffusion	Positive heat capacities and absolute temperature; symmetric nonnegative diffusion tensors.
Optical and drag	$\kappa, C_D, \text{area} \geq 0$; optical fractions in $[0,1]$ sum to one.
Coordinate maps	Positive log arguments and length scales; charts exclude their singular coordinates and use overlaps.
Spectral / orbital axes	Nonzero selected axis projection; nonzero orbital separation.
Spacetime	Positive-definite h, $N > 0$, symmetric K, declared matter, constraints and gauge.

ADM / SCALAR / XD-TAU CLOCK / 43

An exact expanding log-time spacetime

The logarithmic clock admits a nontrivial exact massless-scalar flat FLRW solution. Let τ=t+t_*, T=ln(τ/t_*), t_* > 0, T ≥ 0, and σ ∈ {−1,+1}. On T³ define:

$$N = \tau = t_* e^T, \quad \beta = 0, \quad h_{ij} = e^{2T/3}\delta_{ij}, \quad K_{ij} = -h_{ij}/(3\tau) \tag{H.11}$$

$$\phi = \phi_0 + \frac{\sigma T}{\sqrt{12\pi}}, \quad \Pi = \frac{\sigma}{\sqrt{12\pi}\,\tau}, \quad V = 0 \tag{H.12}$$

$$\rho_E = \frac{1}{24\pi\tau^2}, \quad j_i = 0, \quad S_{ij} = \rho_E h_{ij} \tag{H.13}$$

Direct verification

$$^{(3)}R = 0, \quad K = -1/\tau, \quad K_{ij}K^{ij} = 1/(3\tau^2) \tag{H.14}$$

$$K^2 - K_{ij}K^{ij} = 2/(3\tau^2) = 16\pi\rho_E \tag{H.15}$$

Spatial derivatives vanish, so the momentum constraint is zero. Metric evolution gives ∂_Th_ij=(2/3)h_ij=−2NK_ij. For curvature, S=3ρ_E makes the matter bracket in (H.2b) vanish; both sides of (H.2) are Nh_ij/(9τ²).

$$\partial_T\phi = N\Pi = \sigma/\sqrt{12\pi}, \qquad \partial_T\Pi = -\Pi = NK\Pi \tag{H.16}$$

$$\partial_T N = N = -N^2K, \qquad a^3/N = e^T/(t_* e^T) = 1/t_* \tag{H.17}$$

Thus the scalar equation and harmonic lapse hold; the shift equation is zero. The constant a³/N verifies that T itself is harmonic. Matter conservation $\dot{\rho}_E+3H(\rho_E+p)=0$ follows with $p=\rho_E$ and physical H=1/(3τ). The full solution is defined for T ≥ 0; its past singular limit τ→0 is outside that forward domain.

Formal contribution

The log-time coordinate is connected to an exact expanding spacetime with matter, curvature, constraints and gauge all checked. The construction is an explicit specialization of established Einstein/scalar equations; the coordinate architecture does not imply a new gravitational law.

SOURCE-SCOPE COMPLETION / VERSION 1.5 / 44

Formal overlaps and exact meanings

The original specification leaves primitive formulas, analog-stage composition and update rules as choices. The expansion adds domain names, some described as exact synonyms. This edition supplies concrete realizations and distinguishes mathematical identity from an adapter or a physical model. Historical instructions inside the supplied exports remain source material.

Source term / domain	Formal relationship	Result
Binary-spherical index / DGGS	Hierarchical spherical partition with explicit addresses, area and radius metadata	Constructed index; certification is a separate empirical/standards claim.
Packed hierarchy / succinct tree	Two child-presence bits per actual binary node, with rank/select	Concrete representation; payload and coordinates have additional storage.
Accumulated fields / FRep	A scalar inequality denotes a solid; exact region operations use canonical semantics	Direct representation family; an arbitrary defining field need not be an exact SDF.
Log-polar / hyperbolic browser	Depth-to-radius map versus Poincaré metric and disk automorphisms	Explicit adapters; log-polar coordinates alone are not hyperbolic geometry.
Neural implicit / graph model	A supplied network defines a leaf or a signal decoder	Specialization with parameters; a bit mask alone is not a trained network.
Eulerian hydrocode / plume	Conservative physical fields plus the diagnostic defined below	Physical modeling specialization, with explicit constitutive and flux data.
CRDT / self-reference	Tagged set-union logs and pure replay; snapshot-driven program rewrite	Defined constructions; XOR values themselves are not a monotone join.
Stabilizer / bent function	Graph adjacency generates a graph state; a separate quadratic Boolean function is bent	Constructive bridges; the source four-term XOR remains linear.
ADM / autopoiesis	Metric, curvature, matter and gauge; internally produced shell boundary	Specified physical families, beyond merely naming time slices or feedback.
Temporal folding / connectome / orbit	Gate reuse schedule; neural graph dynamics; controlled orbit/channel maps	Concrete adapters below; empirical device or anatomical identity is not inferred.

Pressure-shell mechanics, scale-patterned heat transport, optical absorption/emission and the complete analog coordinate chain are formal physical or continuous models. Their equations and domains, rather than the names alone, establish the overlap.

FINITE ALGEBRA / INFORMATION CONTENT / 45

Exact bridges: address and parity

$$k = 2^d - 1 + j, \quad d = \lfloor \log_2 (k+1) \rfloor, \quad j = k - (2^d - 1) \tag{J.1}$$

Pad j to d binary digits to recover the unique word w. Thus all finite words, including the empty word, are in bijection with nonnegative integer heap addresses. The sphere cell associated with a word contains many points; it is not in bijection with one point. Coordinate residuals retain that additional information.

At a fixed depth, sorted cell keys j are a total order. A binary interval-search tree splits a key interval at its midpoint; left and right prefixes are its 0/1 extensions. Entity keys (j,ID) resolve multiple occupants. A separately balanced search tree on those keys has logarithmic search depth; arbitrary organizational parent relations need not obey the search order.

$$\mathcal{P}(A,B) = A + B + A^T + B^T \quad (\mathbb{F}_2), \qquad K = N(N-1)/2 \tag{J.2}$$

The image is exactly the symmetric zero-diagonal matrices. For any such Q, take B=0 and A equal to the strict upper triangle of Q. Then P(A,B)=Q, proving surjectivity. The image dimension is K; rank-nullity gives:

$$\dim \ker \mathcal{P} = 2N^2 - K = (3N^2 + N)/2 \tag{J.3}$$

Storing K mask bits is lossless for Q, not for the original pair (A,B). Different preimages collapse to the same mask. If original planes are needed, they remain additional state. This resolves the source's complete-compression claim with an exact information count.

$$|G_Q\rangle = 2^{-N/2} \sum_{z \in \mathbb{F}_2^N} (-1)^{\sum_{i<j} Q_{ij} z_i z_j} |z\rangle \tag{J.4}$$

The fixed labeled adjacency determines the graph-state vector. Conversely its relative amplitudes at z=e_i+e_j recover Q_{ij}, so this is a bijection onto this family of graph states. Its commuting generators were proved in (19.4). Error correction requires a selected code and channel; a one-dimensional graph-state subspace alone supplies no logical-qubit protection claim.

The bent function g(x,y)=x·y on 2k variables has Walsh magnitudes 2^k and 2^(2k−1)−2^(k−1) ones. It is nonlinear and unbalanced. It is an additional operator, not another name for the linear map (J.2).

NEURAL / ANALOG / DIFFERENTIABLE BRIDGES / 46

Canonical graph axis and inverse fitting

For a finite symmetric graph Laplacian L with N≥1, select its smallest strictly positive eigenvalue, or zero if none exists. Let P_λ be the orthogonal projector onto that eigenspace. Select the first canonical basis index j with P_λe_j ≠ 0 and define:

$$v = P_\lambda e_j / |P_\lambda e_j|, \qquad a = X^T v \tag{J.5}$$

A nonempty eigenspace makes that index exist. The projection fixes degeneracy and sign because v_j=|P_λe_j|>0. X contains centered entity positions. Put β=atan2(a_z,a_x) when the xz projection is nonzero, and β=0 for its degenerate case. A tolerance-based numerical eigenvalue grouping records its tolerance separately; it does not change the exact definition.

For the complete three-node graph, L=3I−11ᵀ, P_3=I−11ᵀ/3 and v=(2,−1,−1)/√6. This is a fully determined neural/graph-to-spatial-axis bridge. A connectome application supplies its measured graph and coordinates; the same construction does not assert anatomical equivalence.

An explicit inverse-geometry fit

$$\mathcal{L}(R) = \tfrac{1}{2}(R - R_{\text{target}})^2, \quad R^+ = (1-\eta)R + \eta R_{\text{target}}, \quad 0 < \eta < 1 \tag{J.6}$$

$$R_n - R_{\text{target}} = (1-\eta)^n (R_0 - R_{\text{target}}) \tag{J.7}$$

For positive initial and target radius this convex combination remains positive and converges geometrically. The reference takes R_0=2/3, target 1.7 and η=1/4, checked with exact rational arithmetic. The sampled sigmoid objective (4.1)-(4.2) remains the general inverse-field family with its actual objective derivative; it has no automatic global convergence claim for arbitrary nonconvex targets.

Ray and field evaluation semantics

$$I(s) = I(0)\exp\Big(-\int_0^s \kappa(\xi)\,d\xi\Big), \qquad \kappa \geq 0 \tag{J.8}$$

An integrable supplied path coefficient defines general attenuation. Finite ordered-segment evaluation additionally uses a finite piecewise description, as in the executable profiles. A black-box continuous field may have infinitely many crossings on a bounded ray. Exact arrangement evaluation and bounded-error sampled Klein distance are the two explicit finite geometry realizations; arbitrary black-box topology is not silently treated as a finite algorithm.

EXPLICIT GEOMETRY / INPUT DATA / 47

The retained primitive package

Use local coordinates q=Oᵀ(x−c), with c ∈ R³ and O=R_z(α)R_y(β)R_x(γ) ∈ SO(3). Positive lengths are stored directly or as exp of a dimensionless parameter times a reference length. This defines arbitrary translated/rotated instances of the following leaves.

$$f_{\text{sphere}}(q) = |q| - R, \qquad R > 0 \tag{K.1}$$

$$\mathcal{V} = \{(1,1,1),(1,-1,-1),(-1,1,-1),(-1,-1,1)\} \tag{K.2}$$

$$f_{\text{pyr}}(q) = \max_{v\in\mathcal{V}} \frac{-v\cdot q - L}{\sqrt{3}}, \qquad L > 0 \tag{K.3}$$

The pyramid is the regular tetrahedron conv{Lv : v ∈ V}; each plane contains the other three vertices and excludes the opposite exterior. The sphere field is an exact signed distance. The pyramid max field is a membership field; outside corners it need not equal Euclidean distance.

$$f_{\text{cone}}(q) = \max\left(-q_z,\, q_z - H,\, \frac{\sqrt{q_x^2+q_y^2} + (R/H)q_z - R}{\sqrt{1+(R/H)^2}}\right) \tag{K.4}$$

R,H > 0. Its solid is 0 ≤ q_z ≤ H and √(q_x²+q_y²) ≤ R(1−q_z/H). The cone max field is a finite Lipschitz membership field. All exact CSG nodes import these leaves and derive their canonical regional distance as specified on page 4.

A concrete mask lookup

$$\hat{x} = x/|x| \ (x \neq 0), \quad \hat{x} = e_y \ (x = 0), \quad u = \hat{x}_y, \quad \psi = [\operatorname{atan2}(\hat{x}_z, \hat{x}_x)]_{2\pi} \tag{K.5}$$

Use the owned depth-d equal-area address from page 8, with ψ=0 at poles. Let j_d be its integer cell key. For finite N ≥ 2 and tick n, put:

$$i = j_d \bmod N, \quad j = (i + 1 + (n \bmod N)) \bmod N, \quad m_n(x) = (-1)^{Q_{ij}} \tag{K.6}$$

This supplies the formerly unspecified lookup indices in (5.5), including the origin. It is a piecewise mask on spatial cells. Raw sign-modulated fields are auxiliary; their imports, canonical exact CSG semantics and weak seam derivatives remain those already proved. Mask categories may share many spatial cells without merging entity IDs.

TOPOLOGY / METRIC APPROXIMATION PROOF / 48

An explicit Klein immersion and thickness

$$q = \cos(u/2)\sin v - \sin(u/2)\sin(2v), \quad z = \sin(u/2)\sin v + \cos(u/2)\sin(2v) \tag{K.7}$$

$$\iota(u, v) = c + \ell((a + q)\cos u, (a + q)\sin u, z), \qquad a > 2,\ \ell > 0 \tag{K.8}$$

Identify v with v+2π and (u+2π,v) with (u,−v). Formula (K.8) respects both identifications and defines a smooth map from the compact Klein bottle. This is the figure-eight realization [R30], with parameter names fixed here.

$$|\iota_v|^2 = \ell^2[\cos^2 v + 4\cos^2(2v)] \geq 31\ell^2/64 \tag{K.9}$$

The radial factor a+q > a−2 > 0 gives iota_u a nonzero angular component. Iota_v has only radial/vertical components and is nonzero by (K.9). The derivatives are linearly independent, proving immersion everywhere. The twisted gluing reverses orientation; image self-intersections do not change the abstract surface.

$$D(x) = \min_{u,\, v \in [0,\, 2\pi]} |x - \iota(u, v)|, \quad A_\varepsilon = \{D \leq \varepsilon\}, \quad f_\varepsilon = D - \varepsilon, \quad \varepsilon > 0 \tag{K.10}$$

Compactness gives a global minimum. Moving toward a nearest image point proves {D≤ε}=cl{D<ε}, so the offset is regular closed. Define the raw f_K in (5.5) as f_epsilon=D−ε. It is a membership field; the canonical d_A is the exact signed distance through overlapping thickened sheets.

$$L_u = \ell\sqrt{(a + \sqrt{2})^2 + 1/2}, \quad L_v = \ell\sqrt{5}, \quad \delta = L_u\pi/M + L_v\pi/N \tag{K.11}$$

$$0 \leq D_{M,N}(x) - D(x) \leq \delta \qquad \text{for every } x \tag{K.12}$$

The derivative bounds in (K.11) put a point of the M×N sampled image within δ of every image point, with twisted seam matching. Minimize over the finite sampled image to obtain D_M,N. The triangle inequality proves (K.12) uniformly. This supplies a convergent metric approximation without inferring exact topology from finite samples.

SELF-REFERENCE / TAGGED EVENTS / 49

Finite programs and actual geometry feedback

A finite typed syntax tree has leaves Empty, Full, Sphere, Pyramid, Cone, KleinOffset and rational arrangement regions. Binary nodes are Union, Meet, Difference and SmoothUnion(h). Primitive constants and finite transforms are stored data. Exact nodes evaluate the region algebra; smooth nodes use canonical distance semantics. Programs contain no same-step recursive call to themselves.

$$\mathrm{Replace}(P, \varnothing, T) = T \tag{L.1}$$

A path is a finite 0/1 word. At a binary node descend to its selected child and apply Replace to the remaining path; rebuild the untouched parent. At a leaf with a nonempty remaining path, return the original subtree. Wrap(P,w,op,T) replaces the located subtree S by op(S,T); a missing target is the same identity result. These recursive rules terminate on finite paths and trees.

$$\mathrm{Mat}(O, n) = \mathrm{fold}(\mathrm{Edit}, P_0, \mathrm{sort}\,\{o \in O : \mathrm{epoch}(o) \le n\}) \tag{L.2}$$

Each immutable tag identifies one typed operation and payload. The collision extension first canonicalizes the merged log by its maximum record per tag. Then filter activation epochs and order by (logical version, replica ID, sequence, canonical tie key). Replay from fixed P_0, including late edits. Equal active canonical logs yield equal materialized trees by finite induction.

XOR each distinct tag once. Two tags with the same payload apply twice and cancel; repeated delivery of one tag contributes once. Absolute assignments select the maximum version tuple per field. Physical replicas additionally replay the same activation epochs and source history from the same initial physical state; log convergence alone does not imply identical trajectories.

Geometry changes its stored next program

$$q_n = \mathbf{1}_{A(P_n)}(p_*), \quad b_{n+1} = b_n \oplus q_n, \quad R_{n+1} = 1 + b_{n+1}/4 \tag{L.3}$$

Take a union of the explicit sphere at the origin, cone centered at (4,0,0), and pyramid centered at (−4,0,0). Store cone R=0.75,H=2, pyramid L=0.5, sphere R_n=1+b_n/4 and sample p_*=(1.1,0,0). At $b_0=1$ the sample is inside, so $b_1=0$ and the stored sphere radius changes 1.25→1.0. The next sample is outside. IDs and payloads are copied unchanged.

This closes the original sample→bit→geometry feedback loop. The membrane profile separately supplies physical production of a boundary. A geometry rewrite that changes stored physical energy also records the parameter-work term (C.5); the displayed feedback example changes a geometric program without assigning it unmodeled elastic storage.

ANALOG COORDINATES / ORDERED STAGES / 50

One complete Shark-Geyser composition

Choose 0<r_lo<r<r_hi and theta ∈ R. Length scales r_*,r_0,R_s,ell are positive; $r_c \geq$ ell; $\alpha_g > 0$ has inverse-length units. Periods p_S,p_L and gains b are positive and dimensionless. Set |ε|,|γ|,|a_θ|<1. The unwrapped theta and a third coordinate ζ retain axial turns.

$$T_p(x) = |(x \bmod p) - p/2|, \qquad c_p(x) = T_p(x) - p/4 \tag{M.1}$$

T_p is the source's Shark profile; c_p is its mean-zero version. The earlier centered field T_p−p/2 differs only by a vertical offset. Add the folded modulation to the original log coordinate, so it does not erase the input branch.

$$x = \ln(r/r_*), \quad s_0 = x + \varepsilon c_{p_S}(x), \quad r_S = r_* e^{s_0} \tag{M.2}$$

$$g = r_c + \ell \tanh((r_S - r_c)/\ell), \qquad r_g = r_0 e^{\alpha_g (g - r_c)} \tag{M.3}$$

$$v_- = \ln(r_g(r_{\mathrm{lo}})/r_*), \quad v_+ = \ln(r_g(r_{\mathrm{hi}})/r_*), \quad s = \frac{\ln(r_g/r_*) - v_-}{v_+ - v_-} \tag{M.4}$$

$$t = \mathrm{frac}(\theta/(2\pi)), \quad (\lambda_1, \lambda_2, \lambda_3) = (1 - s, s(1 - t), st) \tag{M.5}$$

Choose triangle vertices (0,0),(1,0),(0,1). The barycentric point is (λ_2,λ_3); its weights are nonnegative and sum to one because 0<s<1 and 0≤t<1. The triangle chart stores the source turn as additional metadata.

$$\mathrm{Len} = s + \gamma c_{p_L}(s), \quad \Theta = \theta + a_\theta \sin(\theta - \beta) \sin(2\pi \mathrm{Len}) \tag{M.6}$$

$$R = R_s e^{b\,\mathrm{Len}}, \quad \mathcal{F}(r, \theta, \zeta) = (R\cos\Theta, R\sin\Theta, \zeta + k\Theta) \tag{M.7}$$

Use k ≠ 0 in length/radian and |ζ|<B<π|k|. Beta is a stored axis phase, or the explicitly selected graph-eigenmode axis from page 18. The sinusoidal phase is a degree-one circle-compatible extension of the source's strip profile. This is a chosen complete realization, with every stage and output specified.

ANALOG MAP / PROOF / 51

Invertibility, turns and material volume

The triangular profile is continuous and 1-Lipschitz. For $x_2>x_1$, $s_0(x_2)-s_0(x_1)\geq(1-|\varepsilon|)(x_2-x_1)>0$. All intervening radial functions are strictly increasing, and $v_+>v_-$. Len differences are similarly bounded below by $(1-|\gamma|)$ times the s difference. Thus R is strictly increasing in r.

$$\partial_\theta\Theta = 1 + a_\theta\cos(\theta-\beta)\sin(2\pi\mathrm{Len}) \geq 1 - |a_\theta| > 0 \tag{M.8}$$

Theta−theta is bounded, so for every fixed r it is an onto strictly increasing map R→R. This proves its unique inverse without choosing an arbitrary angular branch.

$$R = \sqrt{X^2+Y^2}, \quad \mathrm{Len} = \ln(R/R_s)/b, \quad \varphi = \operatorname{atan2}(Y,X) \tag{M.9}$$

$$n = \mathrm{nearestInteger}\left(\tfrac{Z-k\varphi}{2\pi k}\right), \quad \Theta = \varphi + 2\pi n, \quad \zeta = Z - k\Theta \tag{M.10}$$

The nearest integer is unique because $|\zeta/(2\pi k)|<1/2$. Invert the monotone Len map, log normalization, eruption logarithm, chamber artanh, and the monotone Shark-log map in that order; then invert the phase map. Bisection on each declared image interval gives a convergent numerical inverse. For the flat triangle alone, $s=\lambda_2+\lambda_3$ and $t=\lambda_3/s$, with the stored turn supplying theta.

$$\det D\mathcal{F} = R\,R_r\,\Theta_\theta > 0 \tag{M.11}$$

Equation (M.11) holds on open differentiability cells and almost everywhere; the ordinary Jacobian may fail to exist at triangular corners. Equal Cartesian outputs give equal r and phases differing by $2\pi m$. Equal Z gives $\zeta_1-\zeta_2=-2\pi km$. Since $|\zeta_1-\zeta_2|<2B<2\pi|k|$, $m=0$. Strict phase monotonicity proves global injectivity; the inverse is continuous on the image.

For a physical reference body use $\chi_0(r,\theta,\zeta)=(r\cos\theta, r\sin\theta, \zeta+k\theta)$, whose determinant is r and which has the same injectivity proof. The material map is $\chi=F_{map}\circ\chi_0^{-1}$, with deformation gradient $D\,F_{map}(D\chi_0)^{-1}$ and physical volume ratio $R\,R_r\,\Theta_\theta/r>0$ almost everywhere. A chart determinant alone is not a physical volume ratio.

$$c_{p,q}(x) = \tfrac{1}{p}\int_0^p \frac{1-q^2}{1-2q\cos(2\pi t/p)+q^2}\,c_p(x-t)\,dt, \quad 0<q<1 \tag{M.12}$$

This positive normalized periodic Poisson convolution is C∞; its derivative is an average of slopes bounded by one. Replacing triangular functions by (M.12) gives a smooth material profile with the same monotonicity proofs. The tested piecewise map remains the source analog specialization.

STATE DOMAINS / ENERGY / CONTINUATION / 52

A composition theorem for the architecture

A profile consists of finite mode/program data d, continuous domain X_d, initial state, typed coefficient/port maps, ordered stages, source/event schedule, step rule, one total energy H, extensive amounts I, and numerical/encoding semantics. Every right-hand-side quantity is a stored constant, declared input, or output of an earlier operation.

$$\Psi_j : S_j \longrightarrow S_{j+1}, \qquad H(s_{j+1}) - H(s_j) = W_j, \quad I(s_{j+1}) - I(s_j) = J_j \tag{N.1}$$

Finite composition theorem. Assume the event map and each stage/coefficient map are single valued, each stage output lies in the next stage's stated domain, and the final output returns to the next-step domain. Then the ordered update is single valued. On an invariant common domain, every finite iterate is defined. Summing (N.1) proves ΔH=ΣW_j and ΔI=ΣJ_j; shared internal transfers cancel.

Proof: finite induction supplies a unique intermediate state at each stage. Range inclusion supplies the next input. The energy and amount differences telescope. Midpoint invertibility, positive pair/reaction maps, the drag formula and Euler/ALE positivity proofs provide constructive hypotheses for the stated stages. A semidefinite energy is not assumed to bound unmeasured null-space coordinates.

Continuous physical, analog and graph couplings

$$\dot{x}_i = f_i(t, x_i, C_i(t, x), u_i(t)), \qquad x \in X_d \tag{N.2}$$

Inputs are continuous between events and bounded on finite intervals. Require f_i and C_i continuous in time and locally Lipschitz in state uniformly on compact subsets. Picard contraction gives a unique local solution. Prescribed locally finite events with ordered single-valued resets into the next mode's domain give a unique maximal piecewise solution [R31].

A sufficient global continuation condition is a nonnegative proper C¹ function V_d with compact sublevels inside X_d, V̇≤a_T+b_TV between events, and V_after≤α_eV_before+β_e at each of the finitely many resets on [0,T]. Gronwall bounds V, keeps trajectories in a compact subset of their domain, and allows repeated local extension. These are a priori mathematical estimates, not runtime acceptance policies.

Time, nonsmoothness and numerical accuracy

Discrete all-future physical-time coverage additionally requires h_n>0 and Σh_n=∞. Continuity and strict positivity of the step function on compact finite-time state sets suffice. Pointwise Clarke selection need not be locally Lipschitz: L=−|x| gives stationary and delayed departing absolutely continuous a.e. solutions from zero. Its explicit discrete step remains defined; continuous fitting needs separate uniqueness hypotheses.

For the same fixed finite model and source history, uniform stage consistency x+h f_j(x)+O(h²) and stability factor 1+Lh on a compact trajectory neighborhood give first-order composed accuracy by discrete Gronwall. This is separate from continuum spatial convergence.

FINITE DEFORMATION / RECIPROCAL HEATING / 53

A closed thermoelastic constitutive model

On a smooth reference solid Ω_0 let $F=D\chi$, $J=\det F>0$ and $\theta>0$. Prescribe $\rho_0, c_v, k, \mu, \lambda, \theta_0>0$ and thermal coupling b in Pa/K. Use the objective Helmholtz density per reference volume:

$$W_a = \tfrac{\mu}{2}(\operatorname{tr} F^T F - 3 - 2\ln J) + \tfrac{\lambda}{2}(\ln J)^2 - b(\vartheta-\vartheta_0)\ln J \tag{O.1a}$$

$$W = W_a + \rho_0 c_v[\vartheta - \vartheta_0 - \vartheta\ln(\vartheta/\vartheta_0)] \tag{O.1b}$$

The derivatives of this single W determine stress and entropy, rather than prescribing them independently [R32]:

$$P = W_F = \mu(F - F^{-T}) + [\lambda \ln J - b(\vartheta-\vartheta_0)]F^{-T} \tag{O.2}$$

$$\eta = -W_\vartheta = b\ln J + \rho_0 c_v \ln(\vartheta/\vartheta_0), \qquad e_0 = W + \vartheta\eta \tag{O.3}$$

$$Q_0 = -JkF^{-1}F^{-T}\nabla_X\vartheta, \qquad \rho_0\ddot{\chi} = \operatorname{Div} P + \rho_0 b_{\text{ext}} \tag{O.4}$$

$$\rho_0 c_v\dot{\vartheta} = \operatorname{Div}(JkF^{-1}F^{-T}\nabla_X\vartheta) - b\vartheta\operatorname{tr}(F^{-1}\dot{F}) + r_0 \tag{O.5}$$

The reciprocal term in (O.5) follows from $\theta\,\dot{\eta} = -\operatorname{Div} Q_0 + r_0$. Frame indifference holds because $\operatorname{tr}(F^TF)$ and $\det F$ are unchanged by a spatial rigid rotation. The reference conductivity is positive definite for $J>0$. The exact local identities are:

$$\dot{e}_0 = P:\dot{F} + \vartheta\dot{\eta}, \quad \partial_t(\rho_0|\dot{\chi}|^2/2 + e_0) = \operatorname{Div}(P^T\dot{\chi} - Q_0) + \rho_0 b_{\text{ext}}\cdot\dot{\chi} + r_0 \tag{O.6}$$

$$\dot{\eta} + \operatorname{Div}(Q_0/\vartheta) = r_0/\vartheta + \tfrac{Jk}{\vartheta^2}\nabla_X\vartheta\cdot F^{-1}F^{-T}\nabla_X\vartheta \tag{O.7}$$

A fully specified reference takes Ω_0 as the unit ball, $\chi=X$ and $\theta=\theta_0$ on its boundary, $b_{ext}=r_0=0$, and smooth compatible initial displacement/velocity/temperature perturbations supported strictly inside. Their amplitudes lie in a neighborhood of $F=I, \theta=\theta_0$ with positive temperature, injective motion and strong ellipticity. The constitutive identity is valid for every $J,\theta>0$; classical evolution uses its smooth strongly elliptic solution interval.

MECHANICAL AND REGENERATIVE BOUNDARIES / 54

Pressure shells, stability and accretion

For a small-strain isotropic spherical shell with inner/outer radii R_i<R_o, prescribe pressures p_i,p_o and elastic μ>0, bulk modulus K=λ+2μ/3>0. The exact radial equilibrium is:

$$u(r) = Ar + B/r^2, \quad A = \frac{p_i R_i^3 - p_o R_o^3}{3K(R_o^3 - R_i^3)}, \quad B = \frac{(p_i - p_o)R_i^3 R_o^3}{4\mu(R_o^3 - R_i^3)} \tag{O.8}$$

$$\sigma_{rr} = 3KA - 4\mu B/r^3, \qquad \sigma_{\theta\theta} = \sigma_{\varphi\varphi} = 3KA + 2\mu B/r^3 \tag{O.9}$$

Substitution gives dσ_rr/dr+2(σ_rr−σ_θθ)/r=0, σ_rr(R_i)=−p_i and σ_rr(R_o)=−p_o. This closes the differential-pressure/stress meaning of a casing. For finite deformation the corresponding pressure traction is PN=−p_b JF⁻ᵀN on its prescribed loaded boundary; it is not imposed simultaneously with clamping there.

A specified stability model

$$M\ddot{q} + C\dot{q} + (K_0 - pG)q = f, \quad H = \tfrac{1}{2}\dot{q}^T M\dot{q} + \tfrac{1}{2}q^T(K_0 - pG)q \tag{O.10}$$

Take M=diag(2,3,4) kg, C=diag(0.2,0.3,0.4) kg/s, K₀=diag(12,20,35) N/m and G=diag(1,1.5,2) m. Modal displacements q are metres and p is Pa. These declared calibration matrices define a finite shell-mode specialization.

$$p_{\text{crit}} = \min_j (K_{0,jj}/G_{jj}) = 12 \text{ Pa}, \qquad \dot{H} = -\dot{q}^T C\dot{q} + \dot{q}^T f \tag{O.11}$$

For fixed 0≤p<12 Pa and f=0, H is positive definite; its derivative proves stability, and C positive definite gives asymptotic stability. At p=13 Pa the first modal stiffness is negative and its scalar characteristic polynomial has a positive real root. This is the exact threshold of these matrices; a measured real shell supplies its own geometry/material calibration or variational stiffness.

The produced shell is surface accretion

$$\dot{R}_o = \frac{\bar{v}_M V k_d m}{4\pi R_o^2}, \quad \Phi_o(x,t) = |x| - R_o(t), \quad \dot{N}_s = 4\pi R_o^2 \dot{R}_o/\bar{v}_M \tag{O.12}$$

The membrane ODE therefore defines its outer interface speed and exactly accounts for deposited volume. New material is born at the moving outer surface; fixed inner radius remains part of the specification. Increasing N_s is not a mass-preserving deformation of one fixed material reference. The shell chemical/thermal transfer remains (C.14).

ADVECTION / MATERIAL AND SPATIAL DATA / 55

Specified interface motion and fluid coupling

For advection, prescribe Φ_0 and a spatially Lipschitz velocity v. Its characteristic flow χ_t gives $\Phi(x,t)=\Phi_0(\chi_t^{-1}x)$. A complete global reference uses rigid translation and rotation:

$$\Phi_0(X) = \sum_{j=1}^{3} (X_j/a_j)^2 - 1, \quad \chi_t(X) = ut + R_z(\omega t)X \tag{P.1}$$

$$v(x,t) = u + \Omega \times (x - ut), \qquad \Phi(x,t) = \Phi_0(R_z(\omega t)^T(x - ut)) \tag{P.2}$$

Set axes (0.7,1.1,0.9), translation u=(0.23,−0.17,0.11), and Ω=(0,0,0.8) in compatible physical units. The flow is a global diffeomorphism, $\det D\chi=1$, and the chain rule gives $\Phi_t+v\cdot\nabla\Phi=0$. Its ellipsoidal volume $4\pi a_1a_2a_3/3$ is constant. No unspecified V_n is needed for this family.

A declared pressure/curvature interface family

$$V_n = M(\Delta p - \gamma\kappa), \quad n = \nabla\Phi/|\nabla\Phi|, \quad \kappa = \operatorname{div} n \tag{P.3}$$

$$\mathcal{E} = \gamma\,\text{Area} - \Delta p\,\text{Volume}, \quad \dot{\mathcal{E}} = -M\int(\gamma\kappa - \Delta p)^2\,dA \tag{P.4}$$

M>0, γ≥0 and constant Δp are prescribed. Outward normal convention and surface first variation prove (P.4) for smooth closed interfaces, or with vanishing edge work. The level-set PDE is $\Phi_t+V_n|\nabla\Phi|=0$ with supplied initial data; viscosity extensions use their comparison hypotheses [R33]. A scalar solution and a unique sharp surface differ if its zero set fattens.

Actual impermeable fluid-solid interface conditions

$$v_{\text{fluid}} = \dot{\chi}, \quad \sigma_{\text{fluid}}n_{\text{fluid}} + \sigma_{\text{solid}}n_{\text{solid}} = 0, \quad \vartheta_f = \vartheta_s \tag{P.5}$$

$$q_f \cdot n_f + q_s \cdot n_s = 0 \tag{P.6}$$

The common velocity makes mechanical traction powers cancel; equal temperature and opposite heat flux cancel exchanged heat and give consistent entropy flux. These conditions specify an interface adapter between the supplied fluid and solid laws. An accreting/permeable interface instead supplies mass/species flux and transported momentum/enthalpy jump data. The closed shell example prescribes its own accretion law rather than assuming impermeability.

The fixed-grid, ALE, rigid-advection and accretion profiles have their own explicit equations and proofs. A general free-boundary continuum application uses these interface conditions together with its chosen geometry/data and solution theorem; its existence is not deduced from a rendering transform.

ALE FLUX / CONSERVATION / GCL / 56

A positive moving-cell Euler stage

Let time-averaged oriented face data satisfy ∑_f A_fn_f=0 and opposite neighbor signs. Let w_f be normal mesh velocity from swept volume, with V_i⁺=V_i+hΣ_f A_fw_f. Use the ideal-gas admissible cone G from (E.1).

$$\widehat{F}_{ij}^{\mathrm{ALE}} = \frac{F_n(U_i) + F_n(U_j) - w_f(U_i + U_j) - \alpha_f(U_j - U_i)}{2} \tag{P.7}$$

$$\alpha_f = \max_{k=i,j} (|v_k \cdot n_f - w_f| + c_k), \qquad V_i^+ U_i^+ = V_i U_i - h \sum_f A_f \widehat{F}_{ij}^{\mathrm{ALE}} \tag{P.8}$$

$$s_i = V_i - \frac{h}{2} \sum_f A_f(\alpha_f - w_f), \quad B_{jf} = (\alpha_f + w_f) U_j - F_n(U_j) \tag{P.9}$$

$$V_i^+ U_i^+ = s_i U_i + \frac{h}{2} \sum_f A_f B_{jf} \tag{P.10}$$

On the step domain V_i⁺>0 and s_i≥0, put b=α_f+w_f−v_j·n_f≥c_j>0. B has density bρ and internal energy bρe−p²/(2ρb)>0. The admissible states form a convex cone. Equation (P.10) is therefore positive, and division by V_i⁺ proves the new state admissible. Shared opposite face fluxes conserve all extensive quantities.

A concrete swept-geometry constructor

$$\xi_i(t) = i/N + \varepsilon \sin(2\pi i/N) \sin(\omega t), \quad 0 < \varepsilon < 1/(2\pi) \tag{P.11}$$

$$V_i = \xi_{i+1} - \xi_i, \quad w_i = [\xi_i(t+h) - \xi_i(t)]/h \tag{P.12}$$

The periodic one-dimensional reference has fixed endpoints 0 and 1, N=32, ε=0.04, ω=2 and h=0.0005. Since its continuous mesh derivative is at least 1−2πε>0, cells remain ordered. Exact face displacements give the volume change and mesh flux from the same data. A constant U is preserved by (P.8), proving geometric conservation, not merely measuring it.

The verification runs 400 nonlinear steps and records positive s_i, density and internal energy. For general face data the printed step domain is part of the theorem. One direct sufficient step schedule bounds h so V_i+hΣAw>0 and V_i−hΣA(α−w)/2≥0; it changes time resolution and does not reject or alter a proposed state.

The displayed multidimensional algebra assumes consistent integrated geometry; (P.11)-(P.12) supplies an exact moving constructor in one dimension. Spatial convergence and a fully coupled moving continuum theorem require their own hypotheses, as do the fixed-grid continuum approximations.

A CLOSED POSITIVE RADIATION MODEL / 57

Optical scattering, emission and heat

For incoherent gray transport, L(x,ω,t) is nonnegative intensity. Prescribe nonnegative direction-independent coefficients κ_a(x,t),κ_s(x,t), source j≥0, a nonnegative kernel with ∫p(ω′,ω)dω=1, and nonnegative initial/incoming data. Use B(θ)=σ_SBθ⁴/π as the explicit emission law, c>0 and θ≥0.

$$c^{-1}\partial_t L + \omega\cdot\nabla L = -(\kappa_a+\kappa_s)L + \kappa_a B + \kappa_s\int_{S^2} p(\omega',\omega)L(\omega')\,d\omega' + j \tag{Q.1}$$

$$Q_{\mathrm{rad}} = \int_{S^2}\kappa_a(L-B)\,d\omega, \quad u_{\mathrm{rad}} = c^{-1}\int L\,d\omega, \quad F_{\mathrm{rad}} = \int \omega L\,d\omega \tag{Q.2}$$

Integrating (Q.1), normalized scattering cancels exactly: u_rad,t+div F_rad=−Q_rad+∫j. Adding +Q_rad to material energy closes absorption/emission exchange. Pure absorption recovers Beer transport. This is intensity transport with supplied coefficients, rather than an inference of a measured material reflectance.

For a complete periodic transport profile choose x∈T³, continuous nonnegative initial L, continuous coefficients/kernel/source and prescribed θ, all bounded on finite time intervals, with the stated kernel normalization. Characteristics with the absorption integrating factor give a positive Duhamel iteration; bounded scattering gives convergence and uniqueness by Gronwall on successive finite intervals. Thus this prescribed-temperature linear family has a global positive mild solution.

A globally well-defined finite physical family

$$\dot{E} = \sum_j \alpha_j A_j - \beta(E/C)^4 \tag{Q.3}$$

$$\dot{A}_j = -\alpha_j A_j + w_j\beta(E/C)^4 + \sigma\sum_k (P_{jk}-\delta_{jk})A_k \tag{Q.4}$$

E is thermal energy, θ=E/C, C>0. Photon-bin energies A_j≥0; α_j,β,σ≥0; w_j≥0 with Σw_j=1; and P is column-stochastic and nonnegative. These fixed coefficients completely determine the closed cell. The positive orthant is invariant, and E+ΣA_j is constant. Its bounded invariant simplex and locally Lipschitz polynomial rates give a unique global solution.

$$E^+ = \frac{E}{[1+3\beta h E^3/C^4]^{1/3}}, \qquad \Delta A_j = w_j(E-E^+) \tag{Q.5}$$

Equation (Q.5) is exact thermal-emission flow; absorption uses A_j⁺=A_j exp(−α_jh) and adds its loss to E. Scattering uses exp[hσ(P−I)], a positive amount-preserving matrix by the same uniformization proof as (C.6). Their ordered split is positive and energy conserving for every h≥0 in real arithmetic.

For common α set β=α a_rad V, a_rad=4σ_SB/c. Then A_j=w_j a_rad V θ⁴ balances emission and absorption; additionally Pw=w makes scattering stationary. The tested two-bin profile uses its own printed coefficients and symmetric exchange subflow.

PHYSICAL DIAGNOSTIC / SCALE TRANSPORT / 58

Fluid state to Shiitake geometry

A plume envelope now has an explicit map from solved fluid state. Let cell centers X_i, positive volumes V_i/densities ρ_i, temperatures θ_i and ambient θ_a be supplied. Use positive width δ_θ and length ell₀. For a vertical gravity-aligned reference, define:

$$w_i = \rho_i V_i \sigma((\vartheta_i - \vartheta_a)/\delta_\theta), \quad c = \frac{\sum_i w_i X_i}{\sum_i w_i}, \quad z_i = X_{i,z} - c_z \tag{Q.6}$$

$$u_i = w_i \sigma(z_i/\ell_0), \quad c_{\mathrm{cap}} = \frac{\sum_i u_i X_i}{\sum_i u_i}, \quad r_i^2 = |X_{i,xy} - c_{\mathrm{cap},xy}|^2 \tag{Q.7}$$

$$a^2 = \ell_0^2 + \frac{\sum_i u_i r_i^2}{\sum_i u_i}, \quad b^2 = \ell_0^2 + \frac{\sum_i u_i (X_{i,z} - c_{\mathrm{cap},z})^2}{\sum_i u_i} \tag{Q.8}$$

$$r_s = a/3, \quad \ell_s^2 = \ell_0^2 + \frac{\sum_i w_i z_i^2}{\sum_i w_i} \tag{Q.9}$$

Here σ is the logistic function. All denominators and dimensions are positive. Use cap center c_cap and the radii a,b in (13.1); use stem center c, radius r_s and half-height ell_s in (13.2). A changed gravity axis uses a recorded orthonormal frame. This deterministic thermal/mass diagnostic is translation equivariant and supplies actual parameters to the cap/stem field; it does not assume every plume must be toroidal.

A complete scale-patterned heat profile

$$k(r) = k_0 e^{\varepsilon \sin(\omega \ln(r/r_*))}, \qquad \vartheta(r) = \vartheta_b + \frac{P}{4\pi} \int_r^b \frac{ds}{s^2 k(s)} \tag{Q.10}$$

On a≤r≤b with 0<a<b, prescribe k₀,r*,θ_b>0, P≥0, and dimensionless ε,ω. This positive temperature solves steady spherical conduction with total outward heat P, outer boundary θ_b, and inner temperature given by the integral.

$$4\pi r^2 k \vartheta' = -P, \qquad \dot{S}_{\mathrm{gen}} = P[1/\vartheta_b - 1/\vartheta(a)] \geq 0 \tag{Q.11}$$

The derivative proves the differential equation and entropy identity directly. For ω≠0, k(qr)=k(r) with q=exp(2π/|ω|); ω=0 gives k=k₀. This is discrete scale repetition of the coefficient. It does not identify the whole temperature solution or a moving front with a fractal flux limiter.

TRAJECTORY / CONTROL WORK / CHANNEL / 59

A reconfigurable orbital beam profile

For each orbital entity prescribe $R_0>0$, $|\varepsilon|<1$, frequencies ν,ω and phase φ. In a fixed nonrotating plane set $R(t)=R_0[1+\varepsilon \sin(\nu t)]$, $\theta=\omega t+\varphi$, $e_r=(\cos\theta,\sin\theta,0)$, $e_\theta=(-\sin\theta,\cos\theta,0)$.

$$r = Re_r, \quad v = \dot{R}e_r + R\omega e_\theta, \quad \ddot{r} = (\ddot{R} - R\omega^2)e_r + 2\dot{R}\omega e_\theta \tag{R.1}$$

$$a_{\text{control}} = \ddot{r} + \mu_g r/|r|^3, \quad H = m|v|^2/2 - \mu_g m/|r|, \quad \dot{H} = ma_{\text{control}} \cdot v \tag{R.2}$$

This supplies a complete smooth controlled trajectory and its work port, with R always positive. Constant R and $\omega^2=\mu_g/R^3$ give the unforced circular orbit. Nonzero ε reconfigures the formation through the stated control acceleration; the energy change is supplied work, not an unaccounted geometry event.

Geometry-dependent passive channel and beam

$$d_j = |r_j - r_{\text{target}}| > 0, \qquad h_j = \frac{\eta e^{-2\pi i d_j/\lambda}}{d_j \sqrt{\sum_k d_k^{-2}}}, \quad 0 < \eta \le 1 \tag{R.3}$$

$$y = h^T w s, \quad \|w\|^2 = P_{\text{tx}}, \qquad w = \sqrt{P_{\text{tx}}}\,\overline{h}/\|h\| \tag{R.4}$$

Wavelength $\lambda>0$, $P_{tx}\ge 0$ and target position are declared inputs. This normalized scalar channel is passive: $||h||=\eta$. For unit input signal energy, Cauchy–Schwarz proves $|h^T w|^2 \le P_{tx}||h||^2$, and (R.4) attains equality. Received power is $P_{tx}\eta^2$; the remainder belongs to the unreceived radiation channel.

When $\eta=0$, define $h=0$ and $w=\sqrt{P_{tx}}\,e_1$; the output is zero. When $P_{tx}=0$ take $w=0$. These complete the degenerate map. More detailed antenna coefficients replace this declared channel with their own measured/passive operator; the matching theorem then uses that operator.

Each r_j determines its spherical address. For $i\ne j$ define $Q_{ij}=1[d_{ij}\le d_{max}]\,LOS_{ij}$ and $Q_{ii}=0$, with $LOS_{ij}=LOS_{ji}$. The free-space reference takes $LOS=1$. This is a simple undirected coverage graph compatible with the parity/graph adapters. Edges, orbit state and organizational parents remain separate. Channel weights use the same timestamped geometry snapshot.

The reference uses three smooth controlled trajectories, a fixed target, $\eta=0.6$ and $P_{tx}=2$. It checks orbital residuals, the analytic work identity and matched-beam optimality. Signal generation and orbital control consume their declared source ports; the hierarchy supplies identity and lookup.

RELATIVITY / CONSTRAINT EXISTENCE PROOF / 60

Constructive inhomogeneous ADM data

On the flat reference T^3 with periods 2π, choose any constant nonzero symmetric trace-free tensor A_tilde, constant mean curvature $\kappa \neq 0$, arbitrary smooth periodic φ_0, scalar $\Pi_0=0$ and $V=0$. Put $a=|A_tilde|^2/8>0$ and $c=\kappa^2/12>0$.

$$h_{ij} = \psi^4 \delta_{ij}, \qquad K_{ij} = \psi^{-2}\widetilde{A}_{ij} + (\kappa/3)h_{ij} \tag{S.1}$$

$$\Delta\psi + a\psi^{-7} + b(x)\psi - c\psi^5 = 0, \qquad b(x) = \pi|\nabla\phi_0|^2 \tag{S.2}$$

Existence. Choose small $l>0$ with $a\,l^{-7} - c\,l^5 \geq 0$ and large U with $a\,U^{-7} + \|b\|_\infty U - c\,U^5 \leq 0$. They are positive constant sub/supersolutions. For $F(x,s) = a\,s^{-7} + b(x)s - c\,s^5$, choose λ large enough that $\lambda + F_s \geq 0$ on $[l,U]$. The positive inverse of $-\Delta+\lambda$ makes $T(s) = (-\Delta+\lambda)^{-1}[\lambda s + F(x,s)]$ order preserving. Starting at l, its monotone iterates stay between l and U; elliptic regularity gives a positive smooth solution.

$$\Delta u + |\nabla u|^2 + a e^{-8u} + b(x) - c e^{4u} = 0, \qquad u = \ln\psi \tag{S.3}$$

Uniqueness. At a positive maximum of $u_1 - u_2$ their gradients agree and the Laplacian difference is nonpositive. The remaining nonlinearity in (S.3) is strictly decreasing in u, so the difference equation is strictly negative, a contradiction. Interchanging the solutions proves equality.

$$R(h) = -8\psi^{-5}\Delta\psi, \quad |K|_h^2 = \psi^{-12}|\widetilde{A}|^2 + \kappa^2/3, \quad \rho_E = \tfrac{1}{2}\psi^{-4}|\nabla\phi_0|^2 \tag{S.4}$$

Substituting (S.4) into the Hamiltonian constraint gives exactly (S.2). The momentum constraint is zero because A_tilde is divergence free, κ constant and $\Pi_0=0$. Set $N_0=1$, $\beta_0=0$ and the harmonic time derivatives already specified. This supplies a broad inhomogeneous initial-data family with the unique local Einstein–scalar development of (H.1)-(H.10).

The numerical profile uses A_tilde=diag(0.5,−0.5,0), $\kappa=1.5$ and $\varphi_0=0.03\sin x$. Ten plain Newton iterations on periodic grids 32,64,128 yield positive conformal factors and the reported discrete residuals. The analytical existence/uniqueness proof concerns the continuum equation; grid residuals concern the discrete approximation.

HARDWARE / EVALUATION SEMANTICS / 61

Temporal folding and finite precision

A logic network is a finite acyclic graph of total gate functions F_v with nonnegative propagation delays d_v. Registered feedback reads the previous tick's stored state. A folding schedule assigns each gate a start time s_v and reusable hardware unit π_v, with:

$$s_v \geq \max_{u \to v} (s_u + d_u), \qquad [s_v, s_v + d_v) \cap [s_w, s_w + d_w) = \emptyset \quad (\pi_v = \pi_w) \tag{T.1}$$

Give every gate at least one compatible unit and retain intermediate values until consumed. In topological/ID order, assign the earliest available compatible unit after all parents finish; break ties by unit ID. Induction proves the scheduled output equals the unfurled network. Latency is max_v(s_v+d_v). The tested specialization uses synchronous unit-delay rounds and retained registers.

$$Q_{ij} = (A_{ij} \oplus B_{ij}) \oplus (A_{ji} \oplus B_{ji}), \quad i < j \tag{T.2}$$

Each independent mask output uses three binary XOR gates. With K=N(N−1)/2 outputs, one reusable gate executes 3K sequential gate intervals; an unrolled two-layer circuit has latency 2d_XOR at equal delays. The exact tradeoff concerns hardware resources, intermediate registers and elapsed time. The physical RC time constant and gate delay remain in seconds; log-time and angular pinion are coordinate quantities.

Explicit quantization and trajectory error

$$Q_\delta(x)_j = \delta_j \mathrm{roundEven}(x_j/\delta_j), \quad \delta_j > 0, \quad |Q_\delta(x)_j - x_j| \leq \delta_j/2 \tag{T.3}$$

Choose positive rational delta scales and encode those scales and resulting integers exactly. Arbitrary finite magnitudes use variable-length integers. Fixed IEEE formats have finite exponent/mantissa range; replay recovers already rounded values. Neither finite representation losslessly encodes every real state.

$$e_{n+1} \leq (1 + Lh_n)e_n + Ch_n^2 + \varepsilon_n \tag{T.4}$$

For a stable first-order finite update on a compact trajectory neighborhood, epsilon_n bounds its actual quantization/rounding perturbation. Iterating (T.4) gives e_N≤exp(LT)[e₀+CT max h_n+Σ epsilon_n]. This is an accuracy statement when range/conditioning/error bounds are supplied, rather than a claim that snapshot round trips alone prove real-trajectory accuracy.

Locate takes d binary decisions; writing a depth-d word takes at least d output bits. Rank/select navigation has its declared word-RAM/index assumptions. Primitive evaluation cost includes its program size and precision. These mathematical bounds replace source claims of universal instantaneous navigation or complete-state compression; measured device timing would supply a separate empirical record.

VALIDATION RECORD / I / 62

Equation checks: geometry and physics

The accompanying seeded calculation evaluates representative identities and update examples. It records the inputs and residuals for each check. Seed: 20261013; Python/NumPy double precision.

Check	Calculation	Result
Smooth CSG derivative	20,000 scalar pairs; central differences	Max residual 4.46×10^{-10}.
Smooth duality	20,000 pairs	$T_h(a,b) = -S_h(-a,-b)$.
Self-subtraction	20,000 values	$\max(a,-a)$ has empty negative interior.
Empty/full outcomes	Extended distances	Occupancies 0 and 1.
Equal-area profile	20,000 points, depth 14	Area residual $\le1.13\times10^{-17}$.
Cell representative radius	Same point sample	Max error 0.122318 rad; all within cell bounds.
Seam/pole indexing	Four designated points	Deterministic depth-14 addresses.
Parameter fitting	Sphere-radius objective, 12 steps	Loss $0.49 \to 2.32\times10^{-6}$.
Finite-volume exchange	Three internal faces	Total amount $16 \to 16$.
Shell volumes	64 solid-angle cells	Sum equals $4\pi(3^3-2^3)/3$.
Drag power	10,000 relative velocities	Max residual 3.55×10^{-15}.
Closed drag/heat example	$K=5.25$ J initially	$K_{next}=3.967865$ J; heat=1.282135 J.
Coriolis work	10,000 vector pairs	Max absolute residual 2.00×10^{-15}.
Optical slab balance	$I_{in}=7$; $\kappa=1.2$; length=0.7	$I_{out}=3.021974$; absorbed=3.978026.
Port-Hamiltonian identity	1,000 states	Max energy-rate residual 8.88×10^{-16}.
Reaction stoichiometry	Three conserving reaction columns	Total-amount rate residual 9.71×10^{-17}.
Scale-patterned heat	10,000 radii; positive temperatures	Pattern residual 3.11×10^{-15}; conductive entropy ≥0.

The cell-radius row is an additional measurement within the equal-area-profile check. Random samples report observed residuals; exact cancellation, derivative and area identities are established by the equations on their respective pages.

VALIDATION RECORD / II / 63

Equation checks: information and charts

Check	Calculation	Result
Snapshot bit mutation	[0,1,1,0] XOR [1,0,1,0]	[1,1,0,0].
Rank/select topology	Five-node tree with a right-only node	Children and parents recovered from 10 shape bits.
Replica merge	All six orders of three operation sets	One final set; repeated delivery is idempotent.
Graph-state commutation	1,024 random 12-node graphs	$H\Lambda H^T = 0$ over F_2; rank 12.
Bent Boolean spectrum	All 16 inputs and spectral coordinates	Every Walsh magnitude is 4; six one-valued inputs.
Hyperbolic focus	1,000 point pairs	Max distance residual 4.00×10^{-15}.
Periodic phase	1,000 angles; a=0.35	Seam residual 1.78×10^{-15}; derivative ≥0.65.
Log-time chain rule	1,000 time coordinates	dt/dT = t+t_*.
Graph diffusion	Three-node path; signal [3,1,−2]	$x^TLx = 13$; total-signal derivative = 0.
Barycentric log-polar inverse	1,000 radii and angular codes	Radius and turn residuals below 10^{-13}.
Chamber radial inverse	1,000 radii; r_c=1, ell=0.8	Max residual below 10^{-12}.

Recorded result

All 27 scripted checks returned their expected result. They cover the geometry, physical-accounting, reaction, data-structure, distributed-algebra, quantum-graph and coordinate examples identified above.

Reproduction and model definition

The earlier companion script *validate_xdtau_v13.py* writes *validation_results_v13.json*. The v1.5 reference and proof-check files extend this record. The record includes the seed, sample counts, parameters and measured residuals. The edition's equations define the formal adapters; a chosen application profile supplies its material coefficients, observations, sources and discretization.

The finite reference update is reproduced from its serialized rational state, ordered events, model coefficients and scheme. General real-valued semantic state has the explicit quantization relation in the completion chapters. View settings, physical time and random-generator state are recorded wherever they influence the selected profile.

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REFERENCES / II / 65

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[R13] Delpratt, Rahman and Raman. Engineering the LOUDS Succinct Tree Representation.
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[R17] Arnowitt, Deser and Misner. The Dynamics of General Relativity.
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[R18] NASA NAIF. The SPICE Concept.
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https://naif.jpl.nasa.gov/naif/spiceconcept.html

Edition lineage and attribution

Tom Klootwijk's XD-TAU Architectural Paradigm v1.1 and reviewed v1.2 supply the core. The 22-page diff force CSG Tom Klootwijk TAU architecture paradigm export supplies the expanded domain vocabulary. Version 1.3 expanded the domains; v1.4 supplied structural proofs; v1.5 closes the source-derived geometry, analog, event and physical profile definitions.

State evolution is direct. Empty and full-space geometry have explicit semantics. Mathematical domains specify where each representation and equation is defined. Paradigm attribution: Tom Klootwijk. Formal expansion, calculations and document preparation: AI-assisted.

VERSION 1.4 / REPRODUCIBLE RECORD / 66

New verification: proofs and execution

Three new suites supplement the earlier 27 formula examples. General proofs are printed in the completion chapters. Exact finite checks, exact symbolic identities and floating-point examples are reported separately; their counts do not constitute a universal proof of the architecture.

Verification	Evidence and result
CSG: 15 checks	16 exact slab regions; all 256 pairs and 4,096 triples. Boolean laws, zero plateaus, canonical fields, seam and derivative counterexamples pass.
Smooth formulas	20,000 seeded samples. Maximum relative formula residual 6.84×10^{-15}; duality residual zero.
Rational reference: 14 checks	Two integrated steps; 22 stage audits. Exact energy/source and monomer residuals zero; canonical snapshot round trips pass.
IEEE64 reference: 13 checks	64 integrated steps; 832 stage audits. Final energy/source residual 6.22×10^{-15}; maximum stage residual 5.83×10^{-15}.
Nonlinear fluid example	Minimum integrated-run density 0.856965; internal energy density 2.33827; maximum CFL weight 0.6500000000000001.
Other amount balances	Fluid mass 2.22×10^{-16}; momentum 1.67×10^{-16}; chemical amount 8.88×10^{-16} residual.
ADM/scalar: 18 identities	Exact sparse Laurent algebra with rational coefficients: every FLRW identity has zero residual.
Shell proof checks	Exact monomer cancellation, conductance derivative and inward coordinate-face expressions pass.
Shell numerical example	RK4 to t=40 at steps 0.02, 0.01, 0.005; maximum amount drift 5.51×10^{-14}; refinement ratio 16.27.

Seed 20261013; Python 3.12.14, NumPy 2.3.5. Floating residuals are absolute values in the reference's normalized units. The maximum over 120 independent floating midpoint balance tests is 7.11×10^{-14}. An exact forward-Euler regression detects the oscillator energy factor 26/25 at h=1/5.

The rational profile uses first-order backward-Euler pair, reaction and spatial attenuation maps. Its two-step snapshot occupies 65,301 bytes because exact denominators grow. The floating profile uses exponential elementary subflows and first-order coupled splitting. Neither record is a continuum-convergence proof.

REFERENCE PROFILE / PDF ATTACHMENTS / 67

Executable specialization and initial data

The attached code defines two concrete specializations of the composition theorem. The rational specialization omits fluid stages. The floating specialization includes periodic one-dimensional nonlinear flow on eight cells, as a finite physical example within the three-dimensional architectural family. Both retain feedback through chemistry, heat, optical coefficients, geometry and information state.

$$x_i = (i + 1/2)/8, \quad \rho_i = 1 + 0.15\sin(2\pi x_i), \quad v_i = 0.2 + 0.1\cos(2\pi x_i) \tag{I.1}$$

$$p_i = 1 + 0.2\sin(4\pi x_i), \quad \mathcal{E}_i = p_i/0.4 + \rho_i v_i^2/2, \quad \Delta x = 1/8 \tag{I.2}$$

Reference data	Fixed values / laws
Mechanics and medium	q=2/3, p=5/4; body mass 2, spring 3, damping 1/4; medium mass 5, velocity −1/3; quadratic drag coefficient 1/7.
Thermal / fluid	Body energies (4,7), capacities (2,3); γ=1.4, c_v=2.5; viscosity 0.08; direct CFL σ=0.65.
Chemistry	f_i=2+0.1sin(2πx_i), a_i=0.3, b_i=0.2, m_i=0.05; species energy (2,0,0,0).
Feedback	r_i=1+θ_i/10+m_i/100; κ_i=0.25exp[0.1(θ_i−1)]; optical length Δx r_i; k_1=0.12exp[0.02(θ_i−1)], k_2=0.1, k_3=0.05.
Information	Initial neural h_i=0.1sin(2πx_i), decay 0.5, nearest-neighbor tanh coefficient 0.1, temperature drive θ_i/30; initial bits (0,1,1).
Source schedule	Tagged XOR events, force and incident beam power are explicit inputs in run_verification. All gains, pair edges and initial rational data are in the attached source.

The executable specialization uses the listed rational/positive feedback laws rather than assuming every optional decoder in (B.3)-(B.6). Its rendered radii are derived outputs with no additional stored elastic energy; mechanical Q remains fixed. All changed stored energy is booked in the stated physical ledgers.

Embedded reproducibility files

The v1.4 evidence package contains three Python scripts and their three JSON result records, retained as attachments here. v14_reference.py defines the integrated transitions; v14_csg_checks.py checks geometric regressions; v14_adm_shell_checks.py checks spacetime identities and membrane dynamics. Run each with Python and NumPy. Exact arithmetic uses the standard-library Fraction class; no symbolic package is required.

Exact rational snapshots store reduced numerator and positive denominator in canonical hexadecimal. The floating snapshots use float.hex/fromhex. Deterministic replay is verified on the same runtime and evaluation order. General continuum real state remains governed by the semantic/quantization distinction (B.1).

PRIMARY SOURCES / ESTABLISHED THEORY / 68

Completion references

[R22] CGAL. 2D Regularized Boolean Set-Operations.
Regularization and exact geometric implementation. The region proof in this edition is dimension independent.
https://doc.cgal.org/latest/Boolean_set_operations_2/index.html

[R23] Clarke. Generalized Gradients and Applications (1975).
Generalized gradients for locally Lipschitz functions.
https://doi.org/10.1090/S0002-9947-1975-0367131-6

[R24] Hairer. Long-time energy conservation of numerical integrators.
Geometric integration context. Quadratic midpoint invertibility and balance are proved explicitly here.
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[R25] Guermond and Popov. Invariant Domains and First-Order Continuous Finite Element Approximation for Hyperbolic Systems (2016).
Invariant-domain and CFL theory. The displayed Rusanov finite-volume positivity proof is supplied separately.
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[R26] Basarić et al. Error estimates of a finite volume method for the compressible Navier-Stokes-Fourier system, Appendix A.
Periodic smooth-data local NSF existence. Numerical convergence has additional hypotheses.
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[R27] Gourgoulhon. 3+1 Formalism and Bases of Numerical Relativity (2007).
ADM conventions, matter decomposition and gauge equations.
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Hyperbolic reduction, gauge propagation and geometric uniqueness.
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[R29] Rendall. Theorems on Existence and Global Dynamics for the Einstein Equations, §2.5.
Scalar matter and local existence theory.
https://arxiv.org/abs/gr-qc/0505133

The source documents determine the architectural vocabulary and attribution. Their embedded conversation instructions and earlier statements of universal validity are not mathematical evidence. The revised equations, proofs and explicit numerical records determine the claims in this edition.

NEW COMPONENTS / EXACT AND NUMERICAL EVIDENCE / 69

Version 1.5 verification record

Four new executable evidence bundles supplement the preserved v1.3/v1.4 records. General arguments and domains appear with the equations. Exact finite/rational checks and floating computations have distinct meanings.

Area	New evidence
Explicit geometry	17 checks: primitive membership, Klein gluing/immersion, sampled-distance bound, concrete mask lookup and geometry-driven program feedback.
Full analog inverse	1,200 samples with retained turns: radius residual $\le 1.91\times10^{-14}$, theta $\le 1.14\times10^{-13}$, axial residual $\le 4.20\times10^{-14}$.
Analog Jacobian	Determinant identity residual $\le 3.56\times10^{-15}$ on differentiability cells; a thickness-violation regression constructs an actual turn collision.
Events / coupling	24 checks: pure replay, LWW, same-tag repetitions, distinct-tag equal payloads, finite AST geometry, exact parity kernels and gate schedules.
Exact coupled ledger	Six physical ticks and 42 stage audits; final rational energy/source residual exactly zero; positive thermal reservoirs.
Parity / folding	All 262,144 N=3 inputs and all 65,536 N=2 input pairs; N=4 fold schedules agree across 256 sampled inputs.
Thermoelasticity	64 constitutive states: fundamental energy-rate residual 1.78×10^{-15}; derivative finite differences reported separately.
Moving ALE cells	400 steps: mass/momentum/energy residuals $\le 1.78\times10^{-15}$; GCL zero; minimum internal energy density 2.132.
Interfaces / shell	Rigid advection residual 3.55×10^{-15}; shell accretion identity; modal energy residual 2.78×10^{-15} and an unstable p=13 Pa mode.
Inhomogeneous ADM	Positive discrete conformal solutions on 32/64/128 grids; elliptic residuals $\le 2.40\times10^{-13}$; observed refinement ratio 4.0155.
Physical domain bridges	Nine checks: radiation/heat, plume diagnostic, scale conduction, controlled orbits, matched beam and zero-channel cases, spectral axis and exact radius fitting.

The new scripts use Python 3.12.14, NumPy 2.3.5 and standard-library Fraction arithmetic. Analytical proofs establish their stated general identities; integrations, samples, finite differences and grid refinements are separately labeled. No verification assertion modifies an evolution rule.

REQUIREMENTS / CORRECTIONS / REPRODUCTION / 70

Completion against the supplied scope

The source audit records 31 domain/claim rows. Twenty-nine concern the requested architecture, named overlaps or validation of claims; two are unaccepted optional suggestions. The attached closure matrix identifies the source, equations and executable evidence for every row.

Original open choice	Completed definition / evidence
Primitive and Klein realization	K.1-K.12: explicit leaves, transforms, quotient, immersion, thickness and metric approximation.
Mask lookup and core feedback	K.5-K.6 and L.1-L.3: owned cells, typed rewrites and actual sample-driven stored geometry.
Analog composed output	M.1-M.12: entire ordered chain, branch/turn data, inverse, injectivity and physical deformation ratio.
Coupling, events and encoding	L,N,T: activation/version replay, stage domains/ledgers, continuation, rational quantizer and error recurrence.
Physical breadth	O-S: thermoelasticity, pressure-shell modes/accretion, interfaces/ALE, radiation, plume/scale transport, orbital beam and inhomogeneous ADM data.
Exact versus proposed synonyms	J and domain table: identity/isomorphism, constructive adapter and corrected unsupported equality have explicit meanings.
Historical overstated claims	XOR nonlinearity, universal compression/latency, automatic CRDT/quantum protection and coordinate/hardware/brain identity are corrected using the relevant mathematics.

What constitutes the completed mathematical specification

The architecture is a family with explicit state, semantic maps, finite realizations and defined physical adapters. Each theorem quantifies its stated domain: exact finite algebra, unique discrete update, global bounded physical ODE, or local continuum development. Model coefficients, observations and boundary/source functions are declared data. Their being data does not make a defined family an unspecified equation.

The source does not require a theorem that every arbitrary physical coupling is globally smooth. Its valid requirements are fulfilled by the specified constructions and proofs; inaccurate historical equivalences are resolved by validation rather than retained as premises. No physical experiment, standards certificate or universal hardware benchmark is fabricated.

One self-contained PDF

Eighteen attachments preserve scripts/results for v1.3, v1.4 and v1.5, the original scope audit and the closure matrix. Extract and run the Python scripts with NumPy to reproduce their records. The JSON files state parameters, scope and measured residuals. This PDF retains the expanded domain chapters and adds the missing concrete constructions; the physical and analog models are part of the specification.

Additional primary references

[R30] Karcher / UCI Virtual Math Museum. Klein Bottle.
Figure-eight immersion formula. The quotient, immersion bound and sampled-distance proof are supplied in this edition.
https://webapps.math.uci.edu/~vmm/Surface/klein_bottle/klein_bottle.html

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Local existence, uniqueness, continuous dependence and continuation; applied here to the stated finite profile hypotheses.
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[R32] Jiang and Racke. Evolution Equations in Thermoelasticity.
Reciprocal thermoelastic equations and local/continuation theory. The chosen W, its stress/entropy and exact identities are printed here.
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Comparison and viscosity-solution framework for the declared interface PDE families.
https://arxiv.org/abs/math/9207212

[R34] COMSOL. Radiation and Participating Media Interactions.
Propagation, emission, absorption and scattering terms. Finite photon/thermal exchange and global boundedness are derived here.
https://doc.comsol.com/6.4/doc/com.comsol.help.heat/heat_ug_theory.07.064.html

Attribution and edition

Paradigm and original design vocabulary: Tom Klootwijk. Version 1.5 mathematical realizations, derivations, verification and document preparation: AI-assisted. Established mathematics is identified through the references; explicit compatible choices made to close source gaps are presented as this edition’s constructions.

Direct state evolution is retained. Empty/full regions, zero channels, missing rewrite paths and eigenmode degeneracies have defined mathematical results. Positive domains and theorem hypotheses determine which equations apply; they are not runtime clipping, rollback or rejection policies.